Simple Harmonic Motion

Periodic Motion: If a motion on definite path repeats itself after a fixed time interval, then this motion is known as Periodic Motion.

Time Period (T): The min^m time interval after which motion repeats itself is known as time period.

Frequency (f): It is no. of repeatations in 1 sec.

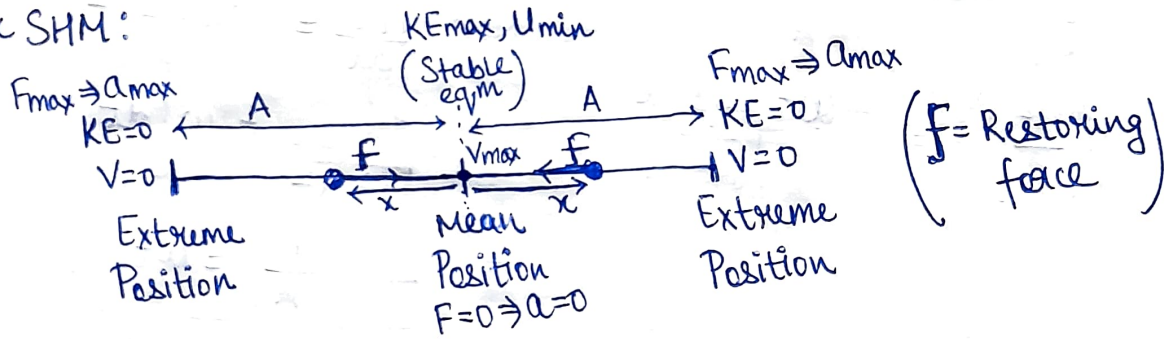
$$\boxed{f = \frac{1}{T}}$$

Unit: ① sec^{-1} = Hertz (Hz)
② CPS (cycles per second)

Oscillatory Motion: Any kind of 'to and fro' motion is known as oscillatory motion.

* If we have TMEC conserved in an oscillatory motion, then that motion is also periodic.

Linear SHM:



① **Amplitude (A):** It is separation b/w equilibrium and extreme position. It is also half of separation b/w extreme position.

Unit = meter Dimension = [L]

Conditions for SHM:

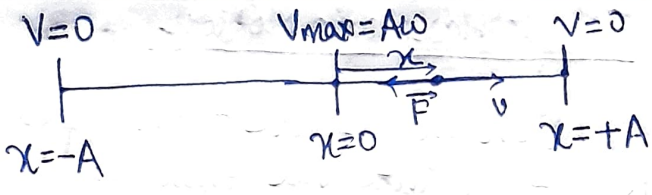
① Restoring force is opposite to displacement from equilibrium position.

② Magnitude of force should be directly proportional to magnitude of displacement.

$$\boxed{\vec{F} = -k\vec{x}}$$

(-ve represents opp. direction)

k = SHM constant
$\vec{F}$ = Restoring Force
$\vec{x}$ = Displacement from mean position.



$F = -kx$

$\Rightarrow ma = -kx$

$a = -\frac{k}{m}x$

$a = -\omega^2 x \quad \left(\text{let } \frac{k}{m} = \omega^2\right)$

$\Rightarrow a = -\omega^2 x$

$\frac{d^2x}{dt^2} = -\omega^2 x$

$\therefore \boxed{\frac{d^2x}{dt^2} + \omega^2 x = 0}$ Differential equation of SHM.

Now, we need to solve this equation.

$a = -\omega^2 x$

$\Rightarrow v\frac{dv}{dx} = -\omega^2 x$

$\int_0^v v\,dv = -\omega^2 \int_A^x x\,dx$

$\Rightarrow \left(\frac{v^2}{2}\right)_0^v = -\omega^2 \left(\frac{x^2}{2}\right)_A^x$

$\Rightarrow v^2 = -\omega^2 (x^2 - A^2)$

$= \boxed{v = \omega\sqrt{A^2 - x^2}}$

If we put $x=0$, $V_{max} = A\omega$

Now, $\frac{dx}{dt} = \omega\sqrt{A^2 - x^2}$

$= \int \frac{dx}{\sqrt{A^2 - x^2}} = \omega \int dt$

$= \sin^{-1}\left(\frac{x}{A}\right) = \omega t + \phi$

$\therefore \boxed{x = A\sin(\omega t + \phi)}$

$\therefore$ Displacement at any time is $x = A\sin(\omega t + \phi)$

* Soln of $\frac{d^2x}{dt^2} + \omega^2 x = 0$ is always $x = A\sin(\omega t + \phi)$

→ $\sin(\omega t + \phi) \in [-1, 1] \Rightarrow x \in [-A, A]$

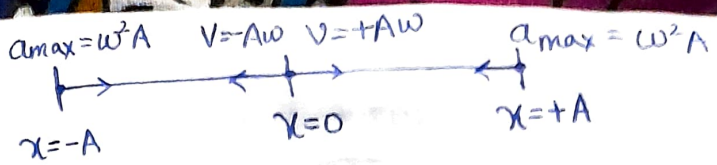
→ $\omega t + \phi$ is called phase. It helps in judgement of position of particle

→ ω is called angular frequency.

$\boxed{\omega = \frac{2\pi}{T} = 2\pi f}$ Unit = Rad/sec

→ ϕ is called phase constant or initial phase.

It tells location of particle at $t = 0s$.



Displacement of particle: $\boxed{x = A\sin(\omega t + \phi)}$

Velocity of particle: $v = \frac{dx}{dt} = A\omega\cos(\omega t + \phi)$

$$\boxed{V = A\omega\cos(\omega t + \phi)}$$

Speed: $|v| = A\omega\sqrt{1 - \sin^2(\omega t + \phi)} = A\omega\sqrt{1 - \frac{x^2}{A^2}} = \omega\sqrt{A^2 - x^2}$

$$\boxed{|V| = \omega\sqrt{A^2 - x^2}}$$

Acceleration of particle: $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

$$\boxed{a = -A\omega^2(\sin\omega t + \phi)}$$

$$\boxed{a = -\omega^2 x} \quad \{x = A\sin(\omega t + \phi)\}$$

$$\boxed{\frac{d^2x}{dt^2} = -\omega^2 x}$$

Draw graph b/w speed v and position x:

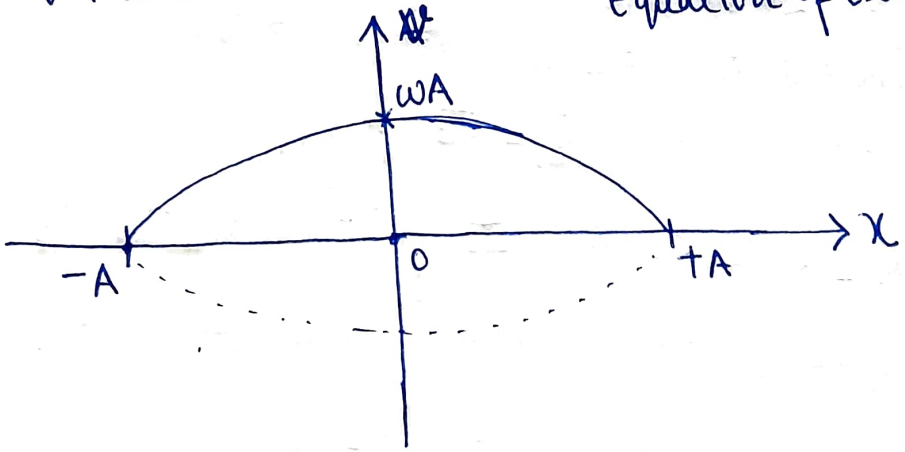
We know, $v^2 = \omega^2(A^2 - x^2)$

$v^2 = \omega^2A^2 - \omega^2x^2$

$v^2 + \omega^2x^2 = \omega^2A^2$

$$\Rightarrow \boxed{\frac{v^2}{\omega^2A^2} + \frac{x^2}{A^2} = 1}$$

Equation of Ellipse.



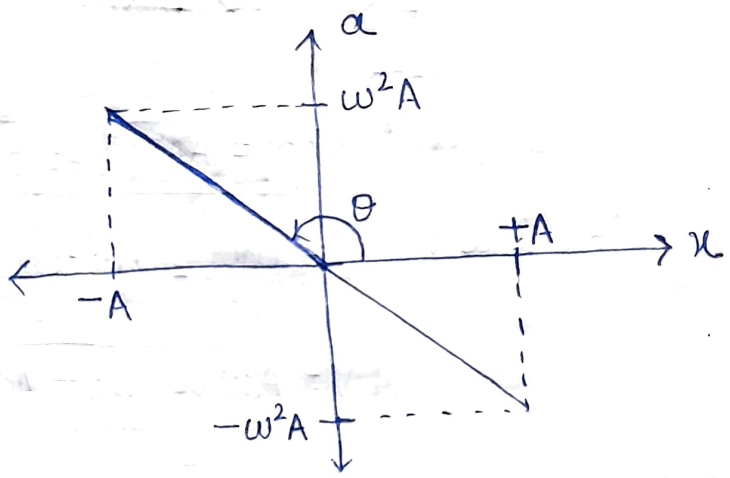
Note: As $|v|$ can't be -ve so lowermost portion of graph ie below x-axis is not considered.

Draw graph b/w acceleration a and position x :

$$a = -\omega^2 x$$

Equation of a straight line.

$$\text{Slope} = \tan\theta = -\omega^2$$



Note :

$$x = A\sin(\omega t + \phi)$$

$$v = A\omega\cos(\omega t + \phi) = A\omega\sin\left(\omega t + \phi + \frac{\pi}{2}\right)$$

$$a = -A\omega^2\sin(\omega t + \phi) = A\omega^2\sin(\omega t + \phi + \pi)$$

→ v is ahead of x by a phase of $\pi/2$.
→ a is ahead of v by a phase of $\pi/2$.
→ x and a have phase difference of π.
(Whenever we have phase difference of π, we say they are out of phase)

Q Given Eqn of SHM ⇒ $x = 2\sin(5\pi t + \pi/3)$. Find the following.

① Amplitude = 2m

② Angular Frequency = $\omega = 5\pi$ rad/s

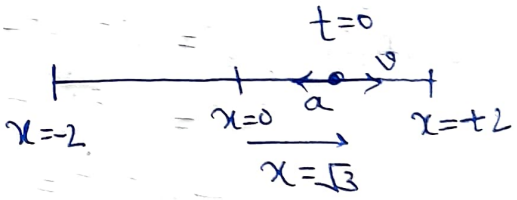
③ Time period = $T = \frac{2\pi}{\omega} = \frac{2\pi}{5\pi} = 0.4$ s

④ Frequency = $f = \frac{1}{T} = 2.5$ Hz

⑤ $V_{max} = A\omega = 2 \times 5\pi = 10\pi$ m/s

⑥ $a_{max} = \omega^2 A = 25\pi^2 \times 2 = 50\pi^2$ m/s²

⑦ Initial position : $x = 2\sin\left(\frac{\pi}{3}\right) = \sqrt{3}$ m
(t=0)



⑧ $v(t) = \frac{dx}{dt} = 2 \times 5\pi\cos\left(5\pi t + \frac{\pi}{3}\right) = 10\pi\cos(5\pi t + \pi/3)$ m/s

⑨ $a(t) = \frac{dv}{dt} = -10\pi \times 5\pi\sin(5\pi t + \pi/3) = -50\pi^2\sin(5\pi t + \pi/3)$ m/s²

⑩ Initial velocity = $10\pi\cos\left(\frac{\pi}{3}\right) = 5\pi$ m/s

⑪ Initial Acceleration = $-50\pi^2\sin(\pi/3) = -25\sqrt{3}\pi^2$ m/s²

Q Given Eqn of SHM ⇒ $x = 10\sin(5t + \pi/6)$. Find the following

(1) $A = 10\ m$

(2) $\omega = 5\ rad/s$

(3) $V_{max} = A\omega = 50\ m/s$

(4) $a_{max} = A\omega^2 = 250\ m/s^2$

(5) Initial position = $x = 5m$

(6) $v(t) = 50\cos(5t + \pi/6)$

(7) $a(t) = -250\sin(5t + \pi/6)$

(8) Initial Velocity = $25\sqrt{3}\ m/s$

(9) Initial Acc^n = $-125\ m/s^2$

(10) V of particle when it is at $x = 6m$

$$V = \omega\sqrt{A^2 - x^2} = 5\sqrt{10^2 - 6^2}$$

$$\boxed{V = 40\ m/s}$$

(11) V at $x = 8m$ ⇒ $V = 5\sqrt{10^2 - 8^2} = 30\ m/s$

(12) a at $x = -5m$ ⇒ $a = -\omega^2 x = -25 \times -5$

$$\boxed{a = 125\ m/s^2}$$

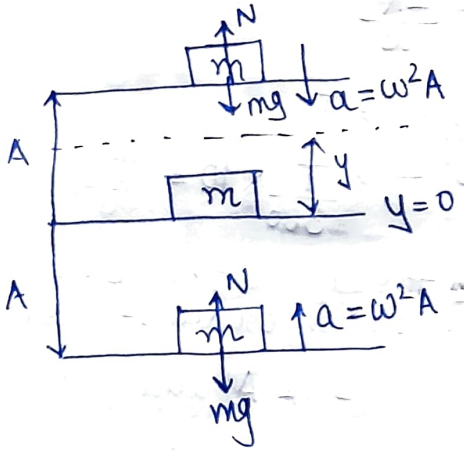
(13) a at $x = 7m$ ⇒ $a = -\omega^2 x = -25 \times 7$

$$\boxed{a = -175\ m/s^2}$$

Q A particle on a planck is performing SHM in vertical direction

Equation ⇒ $y = A\sin\omega t$

Find Normal force on mass m



(1) when planck is at max^m height.

$$mg - N = m\omega^2 A$$

$$\boxed{N = mg - m\omega^2 A}$$

(Least value of N for this SHM)

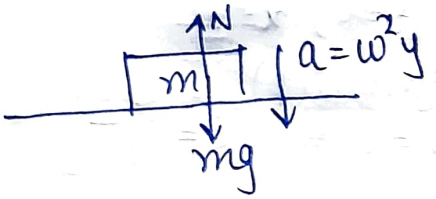
(2) when planck is at max^m depth.

$$N - mg = m\omega^2 A$$

$$\boxed{N = mg + m\omega^2 A}$$

(Max^m value of N for this SHM)

(3) when planck is at y distance above equilibrium position.

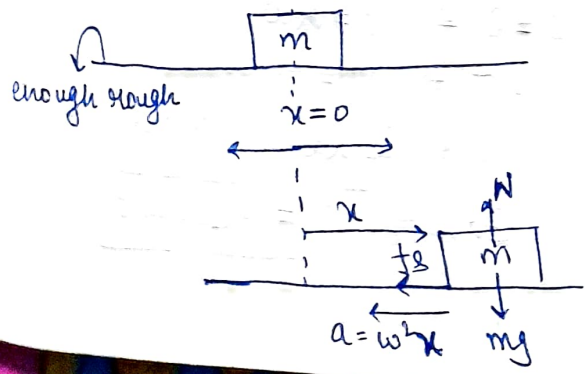


$$mg - N = m\omega^2 y$$

$$\boxed{N = mg - m\omega^2 y}$$

$$\boxed{N = mg - m\omega^2 A\sin(\omega t)}$$

Q



Planck is performing SHM in horizontal

$x = A\sin\omega t$

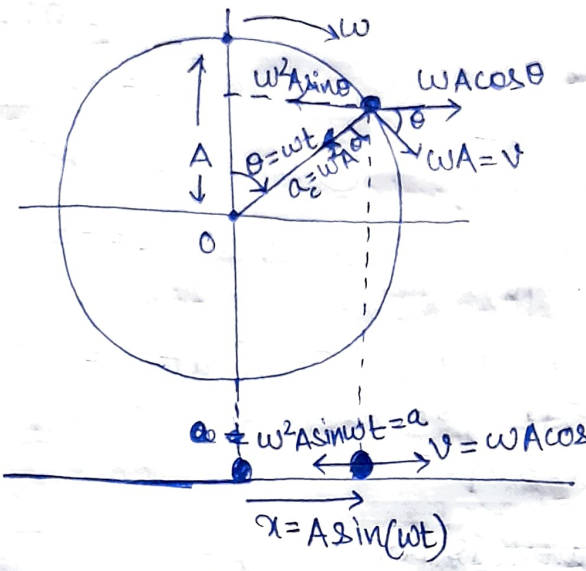
Find value of frictional force at any time t.

$$f_s = m\omega^2 x$$

$$\boxed{f_s = |m\omega^2 A\sin\omega t|}$$

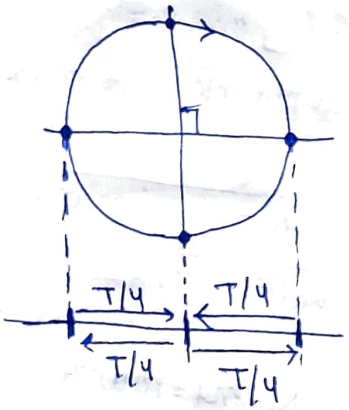
SHM as projection of Uniform Circular Motion (UCM):

* Any SHM can be represented as projection of UCM.



Particle in UCM

Horizontal Planck



Time period $= \frac{2\pi}{\omega}$

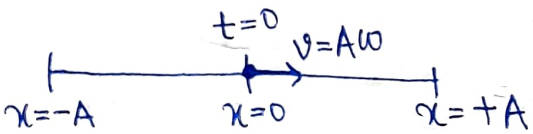
For $360° \longrightarrow T$

$90° \longrightarrow \frac{T}{360} \times 90° = \frac{T}{4}$

∴ Time taken to reach extreme position from mean position or vice-versa $= T/4$.

Q Write down equation of SHM whose amplitude is A and angular frequency ω and its initial position is as shown.

(i)



Method 1: $x = A\sin(\omega t + \phi)$

At $t=0$, $x=0$

$0 = A\sin(\phi)$

$\sin\phi = 0 \Rightarrow \boxed{\phi = 0, \pi}$ $(\phi \in [0, 2\pi]$ (Also $\phi=0$ and $\phi=2\pi$ will give same value)

Now, $\frac{dx}{dt} = v = A\omega\cos(\omega t + \phi)$

At $t=0$, $v = A\omega\cos\phi$

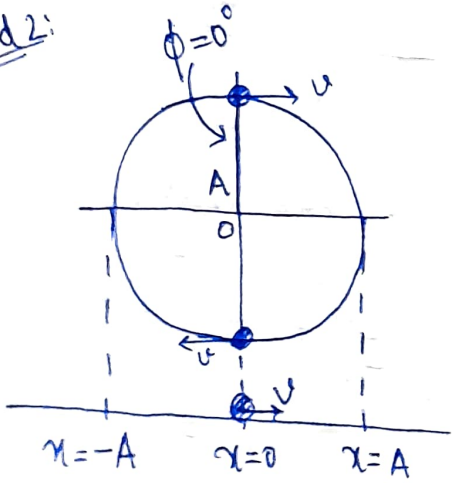
If $\phi = 0 \Rightarrow v = A\omega$

If $\phi = \pi \Rightarrow v = -A\omega$

Acc to fig we have $v = A\omega \Rightarrow \phi = 0$.

Eqn $\Rightarrow \boxed{x = A\sin\omega t}$

Method 2:



∴ Eqn $\Rightarrow \boxed{x = A\sin\omega t}$

(2)

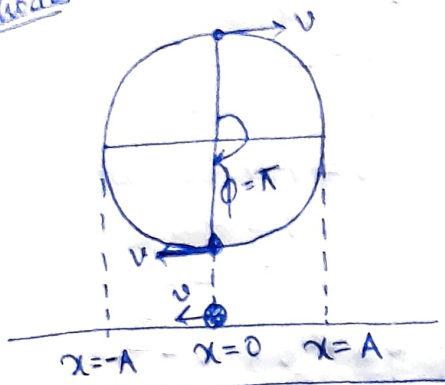
Method 1: $x = A\sin(\omega t + \phi)$

At $t=0$, $x=0$

$0 = A\sin(\phi) \Rightarrow \sin\phi = 0$

$\phi = 0, \pi$

Method 2:



Now, $v = A\omega\cos(\omega t + \phi)$

At $t=0$, $V = A\omega\cos\phi$

If $\phi = 0$, $V = A\omega$

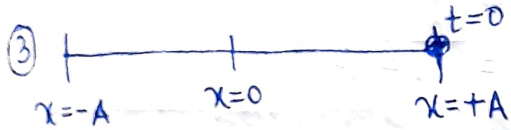
If $\phi = \pi$, $V = -A\omega$

Acc to fig we have $v = -A\omega \Rightarrow \phi = \pi$

Eqn $\Rightarrow$ $x = A\sin(\omega t + \pi) = -A\sin\omega t$

Eqn $\Rightarrow$ $x = A\sin(\omega t + \pi) = -A\sin\omega t$

(3)



Method 1: $x = A\sin(\omega t + \phi)$

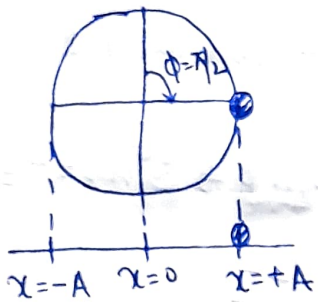
At $t=0$, $x=+A$

$A = A\sin(\phi) \Rightarrow \sin\phi = 1$

$\therefore \phi = \pi/2$

Eqn $\Rightarrow$ $x = A\sin(\omega t + \frac{\pi}{2}) = A\cos\omega t$

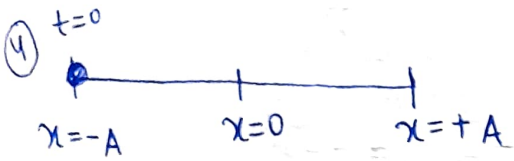
Method 2:



Eqn $\Rightarrow$ $x = A\sin(\omega t + \pi/2)$

$x = A\cos\omega t$

(4)



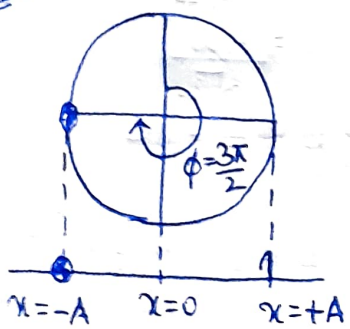
Method 1: $x = A\sin(\omega t + \phi)$

At $t=0$, $x=-A$

$-A = A\sin(\phi) \Rightarrow \sin\phi = -1 \Rightarrow \phi = 3\pi/2$

Eqn $\Rightarrow$ $x = A\sin(\omega t + \frac{3\pi}{2}) = A\sin(\omega t - \frac{\pi}{2}) = -A\cos\omega t$

Method 2:



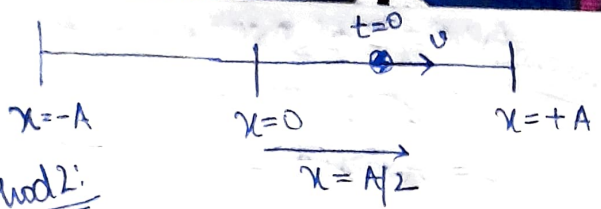
Eqn $\Rightarrow$ $x = A\sin(\omega t + \frac{3\pi}{2}) = A\sin(\omega t - \frac{\pi}{2}) = -A\cos\omega t$

Note: If it said that particle is released from rest then it is at either of the two extremes

If at +ve extreme $\Rightarrow$ $x = A\cos\omega t$

If at -ve extreme $\Rightarrow$ $x = -A\cos\omega t$

(5)



Method 1: $x = A\sin(\omega t + \phi)$

At $t=0$, $x = A/2$

$\frac{A}{2} = A\sin\phi \Rightarrow \sin\phi = \frac{1}{2}$

$$\boxed{\phi = \frac{\pi}{6}, \frac{5\pi}{6}}$$

Now, $v = A\omega\cos(\omega t + \phi)$

At $t=0$, $v = A\omega\cos(\phi)$

If $\phi = \pi/6$, $v = \frac{\sqrt{3}}{2}A\omega$

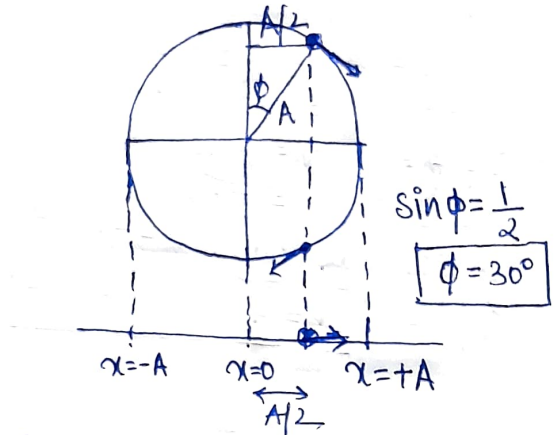
If $\phi = 5\pi/6$, $v = -\frac{\sqrt{3}}{2}A\omega$

But we have $v = +ve$.

$\therefore \phi = \pi/6$.

Eqn = $\boxed{x = A\sin(\omega t + \pi/6)}$

Method 2:

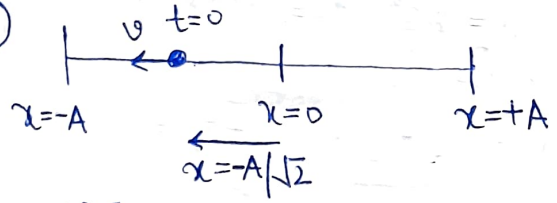


$\sin\phi = \frac{1}{2}$

$\boxed{\phi = 30°}$

Eqn = $\boxed{x = A\sin(\omega t + \pi/6)}$

(6)



Method 1: $x = A\sin(\omega t + \phi)$

At $t=0$, $x = -A/\sqrt{2}$

$\frac{-A}{\sqrt{2}} = A\sin\phi \Rightarrow \sin\phi = -\frac{1}{\sqrt{2}}$

$$\boxed{\phi = \frac{5\pi}{4}, \frac{7\pi}{4}}$$

Now, $v = A\omega\cos(\omega t + \phi)$

At $t=0$, $v = A\omega\cos\phi$

If $\phi = \frac{5\pi}{4} \Rightarrow v = -\frac{A\omega}{\sqrt{2}}$

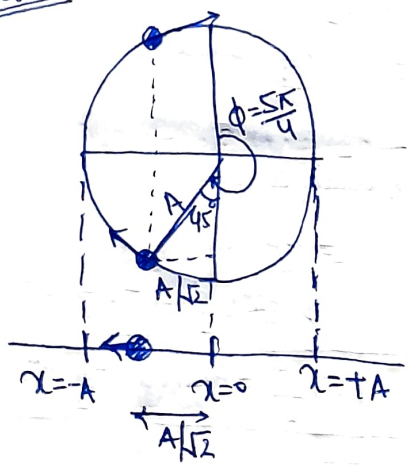
If $\phi = \frac{7\pi}{4} \Rightarrow v = \frac{A\omega}{\sqrt{2}}$

But we have $v = -ve$

$\therefore \phi = 5\pi/4$

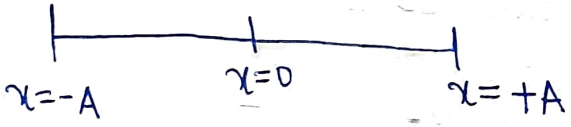
Eqn $\Rightarrow$ $\boxed{x = A\sin(\omega t + \frac{5\pi}{4})}$

Method 2:



Eqn $\Rightarrow$ $\boxed{x = A\sin(\omega t + \frac{5\pi}{4})}$

Q

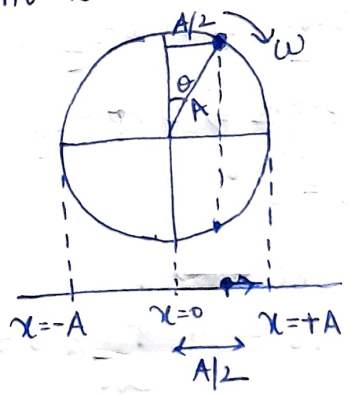


Given, ω

$T = \frac{2\pi}{\omega}$

Find min^m time taken to travel :

① From $x=0$ to $x=A/2$:



$\sin\theta = \frac{A/2}{A} = \frac{1}{2} \Rightarrow \boxed{\theta = \frac{\pi}{6}}$

Also, $\theta = \omega t \Rightarrow t = \frac{\theta}{\omega} = \frac{\pi}{6\omega}$

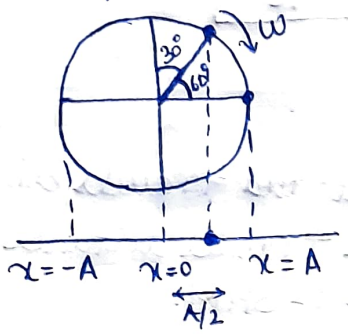
$t = \frac{\pi}{6\omega} = \frac{\pi}{6} \times \frac{T}{2\pi} = \frac{T}{12}$ $\boxed{t = \frac{T}{12}}$

Alternate:

$2\pi \longrightarrow T$

$\pi/6 = \frac{T}{2\pi} \times \frac{\pi}{6} \Rightarrow \frac{T}{12}$

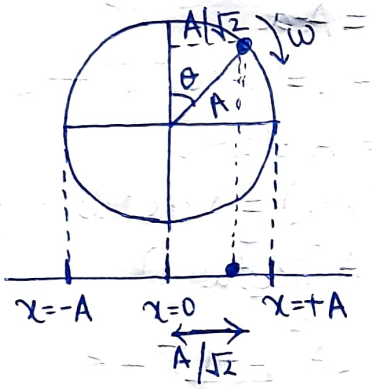
② From $x=A/2$ to $x=A$:



$\theta = \omega t \Rightarrow \frac{\pi}{3} = \omega t \Rightarrow t = \frac{\pi}{3\omega}$

$t = \frac{\pi}{3} \times \frac{T}{2\pi} \Rightarrow \boxed{t = \frac{T}{6}}$

③ From $x=0$ to $x=A/\sqrt{2}$:

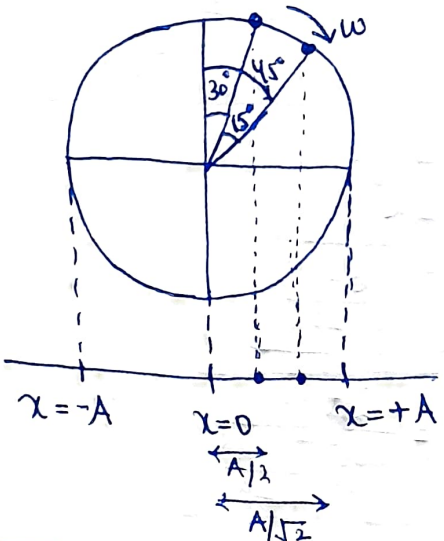


$\sin\theta = \frac{A/\sqrt{2}}{A} \Rightarrow \sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$

Also, $\theta = \omega t \Rightarrow \boxed{t = \frac{\pi}{4\omega}}$

$t = \frac{\pi}{4} \times \frac{T}{2\pi} \Rightarrow \boxed{t = \frac{T}{8}}$

④ From $x=A/2$ to $x=A/\sqrt{2}$:

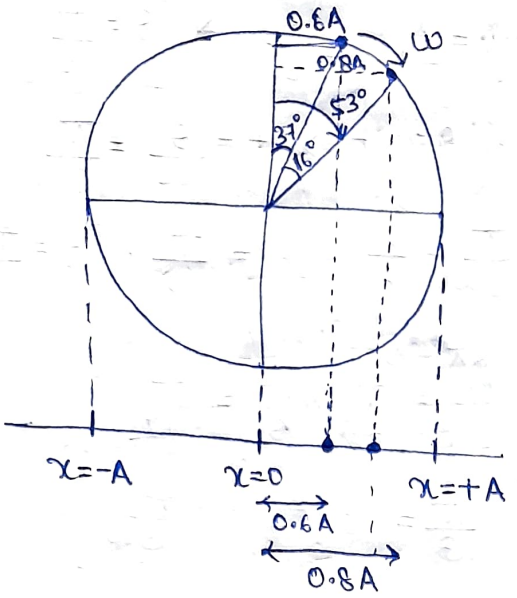


$\theta = \pi/12$

$\theta = \omega t \Rightarrow \frac{\pi}{12} = \omega t \Rightarrow t = \frac{\pi}{12\omega}$

$t = \frac{\pi}{12} \times \frac{T}{2\pi} \Rightarrow \boxed{t = \frac{T}{24}}$

⑤ From $x=0.6A$ to $x=0.8A$



$\sin\theta = \frac{0.6A}{A} \Rightarrow \sin\theta = \frac{3}{5} \Rightarrow \theta = 37°$

$\sin\phi = \frac{0.8A}{A} \Rightarrow \sin\phi = \frac{4}{5} \Rightarrow \phi = 53°$

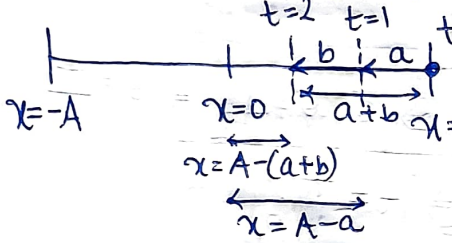
$\alpha = \phi - \theta = 16°$

$360 \longrightarrow T$

$16 \longrightarrow \frac{T}{360} \times 16 = \frac{2T}{45}$

$$\boxed{t = \frac{2T}{45}}$$

Q A particle performing SHM starts from rest and travels distance 'a' in 1 sec. In next 1 second particle travels distance 'b' with increasing speed. Find amplitude of SHM.



We know, $x = A\cos\omega t$

At $t=1$ $\quad x = A-a$

$A - a = A\cos\omega$ — ①

At $t=2$ $\quad x = A-(a+b)$

$A-(a+b) = A\cos 2\omega$ — ②

$\Rightarrow A-(a+b) = A(2\cos^2\omega - 1)$

$A-(a+b) = 2A\left(\frac{A-a}{A}\right)^2 - A$

$A-(a+b) = 2\frac{(A^2+a^2-2Aa)}{A} - A$

$\Rightarrow A-(a+b) = \frac{2A^2+2a^2-4Aa-A^2}{A}$

$= A^2 - Aa - Ab = A^2 + 2a^2 - 4Aa$

$3Aa - Ab = 2a^2$

$$\boxed{A = \frac{2a^2}{3a-b}}$$

Find Time period of this SHM.

Put A in ①

$$= \frac{\frac{2a^2}{3a-b} - a}{\frac{2a^2}{3a-b}} = \cos\omega$$

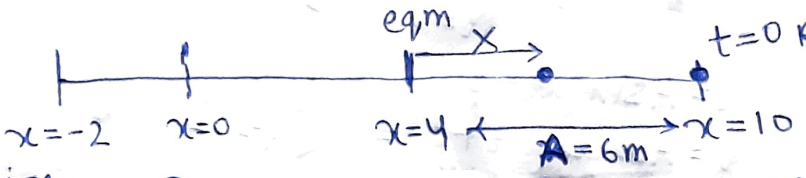
$$= \frac{-a^2+ab}{2a^2} = \cos\omega$$

$\Rightarrow \cos\omega = \frac{b-a}{2a}$

$$\boxed{\omega = \cos^{-1}\left(\frac{b-a}{2a}\right)}$$

$$\boxed{T = \frac{2\pi}{\omega} = \frac{2\pi}{\cos^{-1}\left(\frac{b-a}{2a}\right)}}$$

Q A particle of mass 500gm is released from $x = 10m$ and it experiences a variable force $F = (8 - 2x)$ Newton. $x \equiv$ x-coordinate from origin



① Find eqm position.

At eqm $\Rightarrow F = 0$

$8 - 2x = 0$

$\boxed{x = 4m}$

② Find amplitude of oscillatory motion.

$A =$ Sep b/w eqm and extreme

$A = 10 - 4 \Rightarrow \boxed{A = 6m}$

③ Find min^m value of x-coordinate

$x = 4 - 6$

$\boxed{x = -2}$

④ Prove that it will perform SHM.

Let us assume particle is at 'X' from eqm position.

$F = 8 - 2x = 8 - 2(4 + X)$

$\boxed{\vec{F} = -2X}$

As It is of type $\vec{F} = -kx$

It is SHM.

⑤ Find angular frequency and time period of SHM.

$F = 2X = ma$

$2X = 0.5a$

$\boxed{a = 4X}$

In SHM, $a = \omega^2 x$

By comparing $\omega = 2 rad/s$

$T = \frac{2\pi}{\omega} = \pi sec$

Note: If we have force of type $F = a - bx$, we can directly find acc^n by this eqn. $\boxed{-bx = ma}$

★ Concept: Constant Force have no effect on ω, f and T.

⑥ Find Vmax

$V_{max} = A\omega = 6 \times 2$

$\boxed{V_{max} = 12 m/s}$

⑦ Find KE_{max} or Energy of SHM or Energy of Oscillation.

$KE_{max} = \frac{1}{2} m V_{max}^2 = \frac{1}{2} \times 0.5 \times 12^2$

$\boxed{KE_{max} = 36 J}$

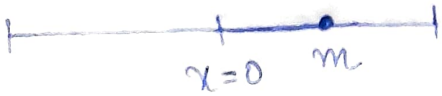
⑧ Velocity of particle when it crosses origin.

$v = \omega\sqrt{A^2 - x^2}$

$v = 2\sqrt{6^2 - 4^2} = 2\sqrt{20}\ m/s$

$\boxed{v = 2\sqrt{20}\ m/s}$

Energy of SHM:



$x = A\sin(\omega t+\phi)$

$v = A\omega\cos(\omega t+\phi) = \omega\sqrt{A^2-x^2}$

① Kinetic Energy:-

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2\cos^2(\omega t+\phi) = \frac{1}{2}m\omega^2(A^2-x^2)$$

$\cos^2 x \in [0,1] \Rightarrow KE \in [0, \frac{1}{2}m\omega^2A^2]$ $\quad KE_{max}$ at $x=0$

Now, $KE = \frac{1}{2}mA^2\omega^2\cos^2(\omega t+\phi) = \frac{1}{2}mA^2\omega^2\left\{\frac{1+\cos(2\omega t+2\phi)}{2}\right\}$

$$KE = \frac{1}{4}m\omega^2A^2 + \frac{1}{4}m\omega^2A^2\cos(2\omega t+2\phi)$$

$\therefore$

	SHM	KE
Angular freq	ω	2ω
Time period	T	T/2
Frequency	f	2f

② Potential Energy:-

$a = -\omega^2x \Rightarrow F = -m\omega^2x$

We know, $F = -\frac{dU}{dx} \Rightarrow -\frac{dU}{dx} = m\omega^2x$

$$\int_0^U dU = m\omega^2\int_0^x x\,dx$$

$$U = \frac{1}{2}m\omega^2x^2$$

$$= \frac{1}{2}m\omega^2A^2\sin^2(\omega t+\phi)$$

$\Rightarrow \sin^2 x \in [0,1] \Rightarrow U \in [0, \frac{1}{2}m\omega^2A^2]$

U_{min} at $x=0$

$$U = \frac{1}{2}m\omega^2x^2$$

We know $\omega^2 = k/m$

$$U = \frac{1}{2}kx^2$$

k → SHM constant

$$= \frac{1}{2}m\omega^2A^2\left\{\frac{1-\cos(2\omega+2\phi)}{2}\right\} = \frac{1}{4}m\omega^2A^2 - \frac{1}{4}m\omega^2A^2\cos(2\omega t+2\phi)$$

$\therefore$

	SHM	PE
Angular frequency	ω	2ω
Time Period	T	T/2
Frequency	f	2f

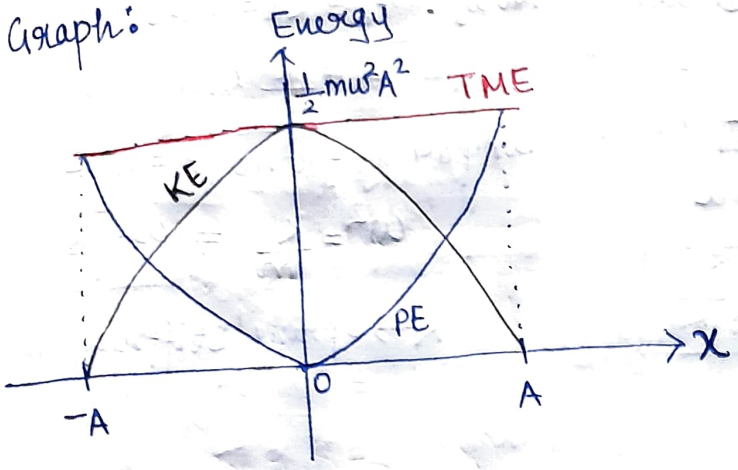
③ Total Energy:

$TE = KE + PE = \frac{1}{2}m\omega^2(A^2 - x^2) + \frac{1}{2}m\omega^2x^2$

$$\boxed{TE = \frac{1}{2}m\omega^2A^2}$$

$\therefore$ TME is conserved.

Graph:



$U = \frac{1}{2}m\omega^2x^2$

$KE = \frac{1}{2}m\omega^2(A^2 - x^2)$

Note: Given $f(t)$, $t : t_1 \to t_2$

$$F_{avg} = \frac{\int_{t_1}^{t_2} f(t)\,dt}{\int_{t_1}^{t_2} dt}$$

$$\therefore \boxed{F_{avg} = \frac{\int_{t_1}^{t_2} f(t)\,dt}{t_2 - t_1}}$$

Q A particle of mass m is performing SHM with eqn $x = A\sin(\omega t)$.
Find ① $(PE)_{avg}$ wrt x if x varies from $x = 0$ to $x = +A$.

$$(PE)_{avg} = \frac{\int_0^A U\,dx}{\int_0^A dx} = \frac{\frac{1}{2}m\omega^2\int_0^A x^2\,dx}{A} = \frac{\frac{1}{2}m\omega^2\left(\frac{A^3}{3}\right)}{A} = \frac{1}{6}m\omega^2A^2$$

② $(PE)_{avg}$ wrt t if t varies from $t = 0$ $t = T/4$.

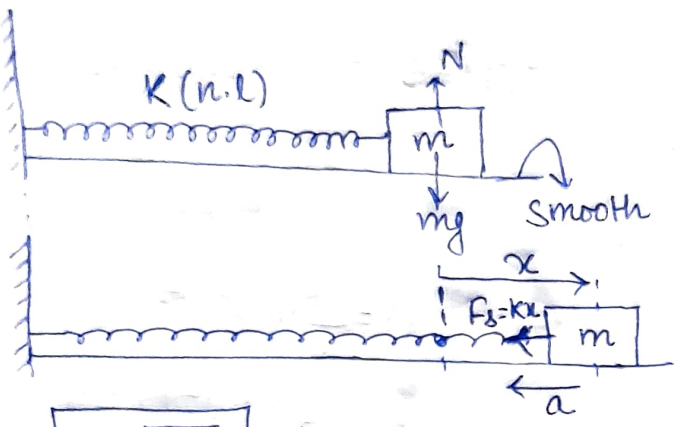
$$(PE)_{avg} = \frac{\int_0^{T/4} U\,dt}{\int_0^{T/4} dt} = \frac{\frac{1}{2}m\omega^2\int_0^{T/4} A^2\sin^2\omega t\,dt}{T/4} = \frac{\frac{1}{2}m\omega^2A^2}{T/4}\int_0^{T/4}\left(\frac{1-\cos 2\omega t}{2}\right)dt$$

$$= \frac{\frac{1}{2}m\omega^2A^2}{2\times\frac{T}{4}}\left(t - \frac{\sin 2\omega t}{2\omega}\right)_0^{T/4} = \frac{\frac{1}{2}m\omega^2A^2}{2\times\left(\frac{T}{4}\right)}\left\{\frac{T}{4} - 0\right\}$$

$\left\{\frac{\sin 2\omega\left(\frac{T}{4}\right)}{2\omega} - \frac{\sin\frac{\omega}{2}\times\frac{2\pi}{\omega}}{} = \sin\pi = 0\right\}$ ($T = \frac{2\pi}{\omega}$)

$$= \frac{1}{4}m\omega^2A^2$$

Spring-Mass System:



$F_{spring} = F_{Restoring} = kx$

As $F \propto x$ and F acts opposite to x. It is a SHM.

Now, $F_s = kx = ma$

$$a = \frac{k}{m} x$$

Comparing from $a = \omega^2 x$

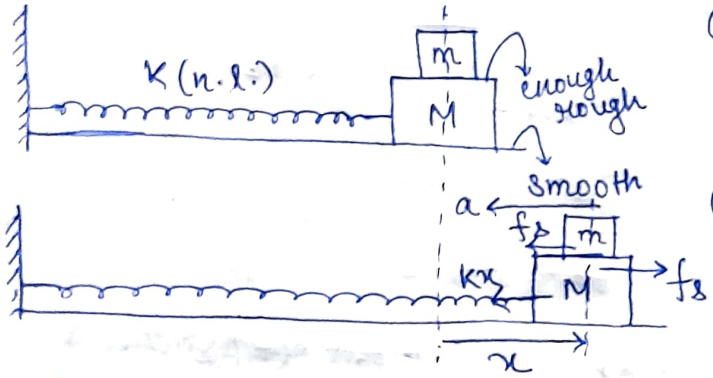
$$\omega^2 = \frac{k}{m} \Rightarrow \boxed{\omega = \sqrt{\frac{k}{m}}}$$

k → Spring / Stiffness constant

$$\therefore \boxed{\omega = \sqrt{\frac{k}{m}}} \qquad \boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}}$$

$$\boxed{f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}}$$

Q



① If M+m is displaced slightly along horizontal dirⁿ, find ω, T and f of their SHM.

② Find values of f_s acting b/w blocks when they are at distance 'x' from equilibrium position.

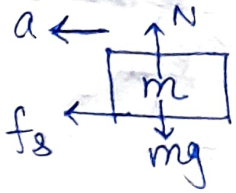
① For (M+m) f_s is internal force.

using NL on (M+m) ⇒ $kx = (M+m)a$

$$a = \frac{k}{m+M} x$$

Also $a = \omega^2 x \Rightarrow \boxed{\omega = \sqrt{\frac{k}{M+m}}}$

$$\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{M+m}{k}}} \qquad \boxed{f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{M+m}}}$$

② 

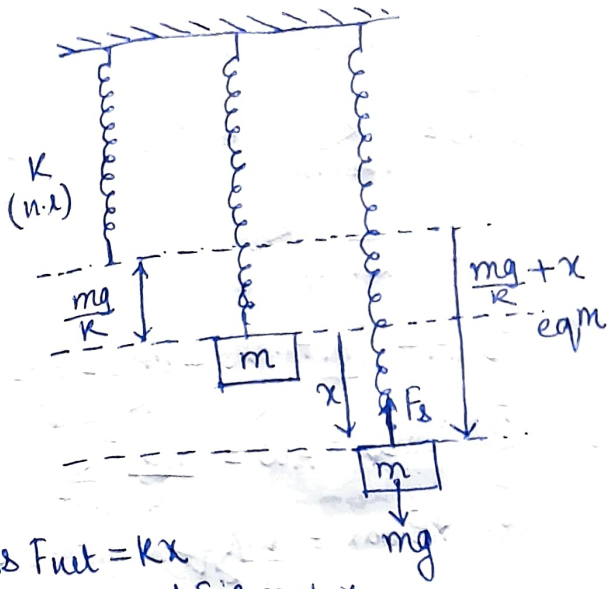
$f_s = ma$

$$\boxed{f_s = m\frac{kx}{m+M}}$$

∴ We can directly use the results, ① $\omega = \sqrt{\frac{k}{m}}$, ② $T = 2\pi\sqrt{\frac{m}{k}}$

③ $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ for these type of situations.

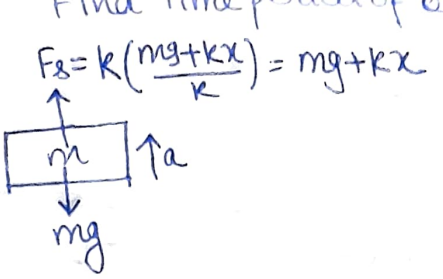
Q



As $F_{net} = Kx$

$\Rightarrow F \propto x$ and F is opp to x

$\rightarrow$ It is SHM.



$\rightarrow mg + kx - mg = ma$

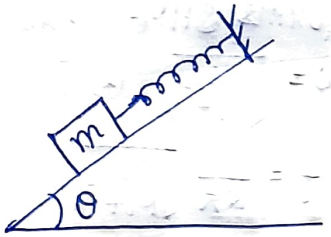
$kx = ma$

$$a = \frac{k}{m}x \Rightarrow \omega = \sqrt{\frac{k}{m}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

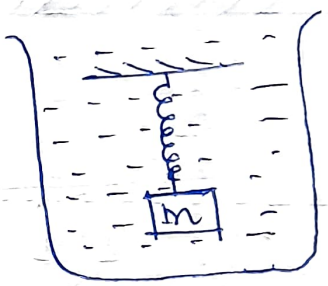
* We know that constant forces have no effect on T. So, even if we had mg acting downwards, T comes out to be same. The role of mg is just to shift the equilibrium position.

Ex



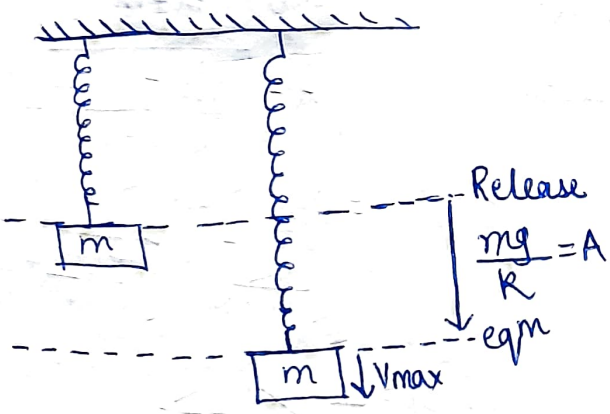
$$T = 2\pi\sqrt{\frac{m}{K}}$$

Ex



$$T = 2\pi\sqrt{\frac{m}{K}}$$

Q



Here, at release position, $V = 0$

∴ Release position is extreme pos.

V_{max} will be at eqm.

A = Sep b/w extreme & eqm $= \frac{mg}{K}$

① Find max speed of the block

② Find $(KE)_{max}$ of the block

① $\omega = \sqrt{\frac{k}{m}}$ $\quad V_{max} = A\omega$

$A = \frac{mg}{k}$ $\quad = \frac{mg}{k}\left(\sqrt{\frac{k}{m}}\right)$

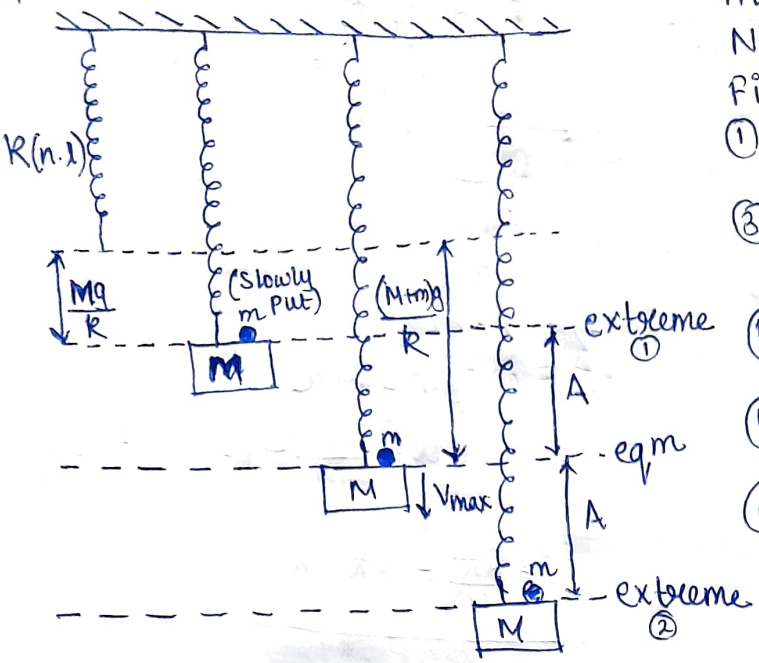
$$V_{max} = g\sqrt{\frac{m}{k}}$$

② $(KE)_{max} = \frac{1}{2}m v_{max}^2$

$= \frac{1}{2} m g^2 \frac{m}{k}$

$$(KE)_m = \frac{m^2 g^2}{2k}$$

Q



(Initially M was at eqm at extreme ① but when we put m on it, eqm position shifts down)

m is placed slightly on M. Now, (M+m) are performing SHM. Find.

① $\omega = \sqrt{\frac{k}{M+m}}$ ② $T = 2\pi\sqrt{\frac{M+m}{k}}$

③ $f = \frac{1}{2\pi}\sqrt{\frac{k}{M+m}}$

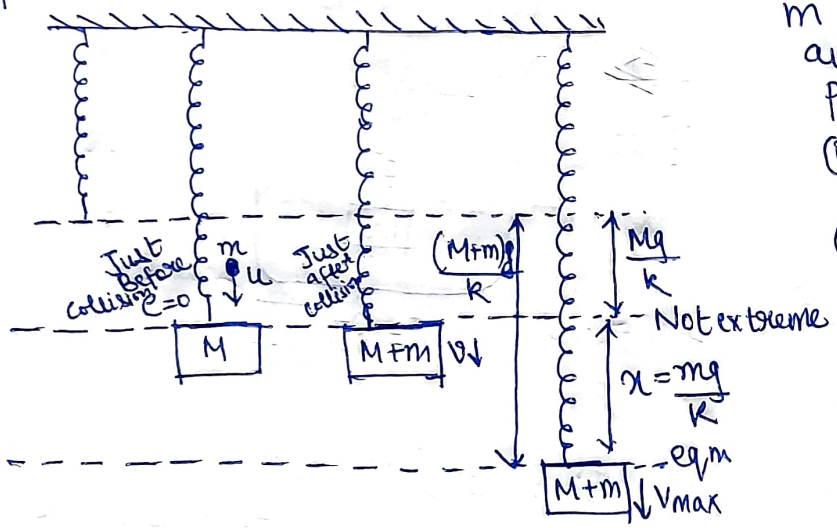
④ $A = \frac{(M+m)g}{k} - \frac{Mg}{k} = \frac{mg}{k}$

⑤ $V_{max} = A\omega = \frac{mg}{k}\sqrt{\frac{k}{M+m}}$

⑥ $KE_{max} = \frac{1}{2}(M+m)\left(\frac{m^2g^2}{k^2}\cdot\frac{k}{(M+m)}\right)$

$= \frac{m^2g^2}{2k}$

Q



m strikes M with speed u and sticks to it. Now (M+m) performs SHM. Find.

① $\omega = \sqrt{\frac{k}{M+m}}$ ② $T = 2\pi\sqrt{\frac{M+m}{k}}$

③ $f = \frac{1}{2\pi}\sqrt{\frac{k}{M+m}}$

④ $A \Rightarrow v = \omega\sqrt{A^2 - x^2}$

$\frac{m^2u^2}{(M+m)^2} = \frac{k}{M+m}\left(A^2 - \frac{m^2g^2}{k^2}\right)$

$A = \sqrt{\frac{m^2u^2}{k(M+m)} + \frac{m^2g^2}{k^2}}$

Using M/C on M+m, before & after collision

$P_i = P_f$

$mu + 0 = (M+m)v$

$$v = \frac{mu}{M+m}$$

As $v \neq 0 \Rightarrow$ Initial position is not extreme position of SHM.

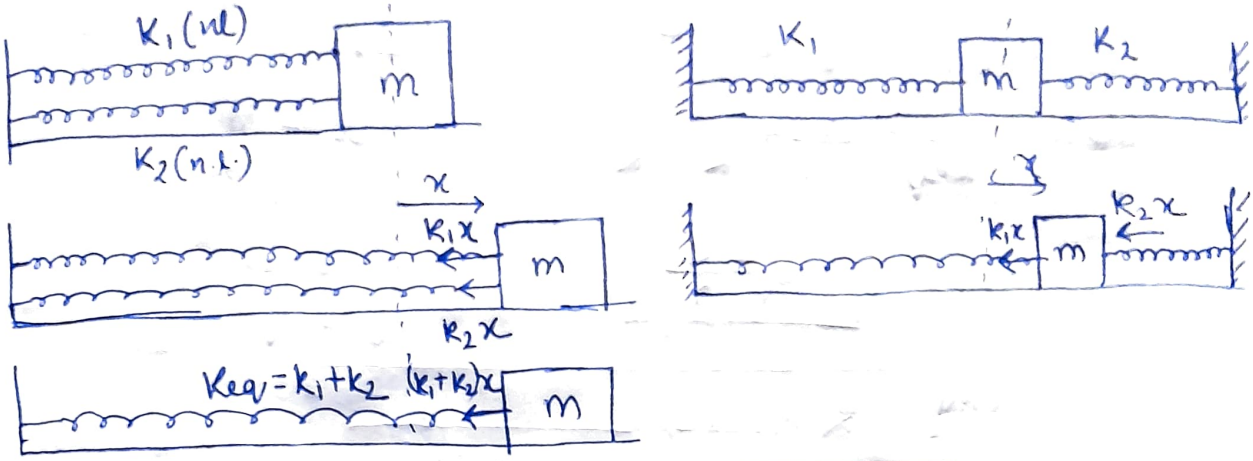
⑤ $V_{max} = A\omega$

$= \sqrt{\frac{k}{m+M}}\left(\sqrt{\frac{m^2u^2}{k(M+m)} + \frac{m^2g^2}{k^2}}\right)$

⑥ $KE_{max} = \frac{1}{2}(M+m)\frac{k}{M+m}\left\{\frac{m^2u^2}{k(M+m)} + \frac{m^2g^2}{k^2}\right\}$

$= \frac{k}{2}\left\{\frac{m^2u^2}{k(M+m)} + \frac{m^2g^2}{k^2}\right\}$

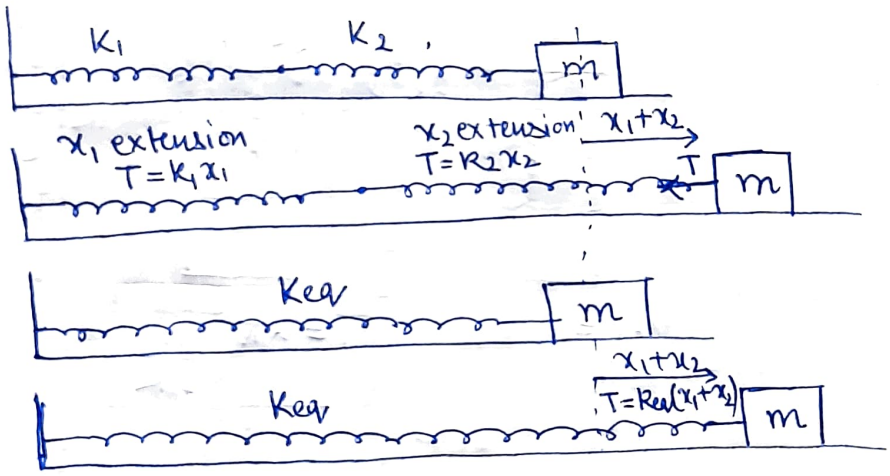
(I) Parallel Combination



$\therefore$ In parallel $\boxed{K_{eq} = K_1 + K_2}$

(II) Series Combination.

↳ Springs connected end to end
↳ Same Tension.



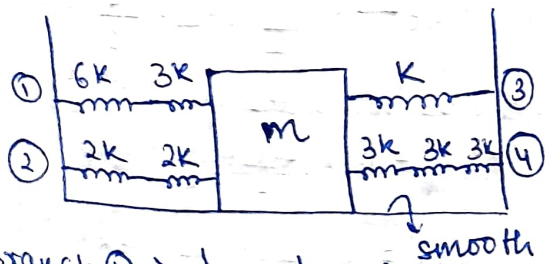
$T = K_1x_1 = K_2x_2$

$\boxed{x_1 = \frac{T}{K_1}}$ $\boxed{x_2 = \frac{T}{K_2}}$

$\frac{T}{K_{eq}} = \frac{T}{K_1} + \frac{T}{K_2}$

$\therefore$ In series $\boxed{\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2}}$

Q



Find Time period for small oscillations of block.

Branch ① $\Rightarrow \frac{1}{K_{eq}} = \frac{1}{6K} + \frac{1}{3K} \Rightarrow K_{eq} = 2K$

Branch ② $\Rightarrow \frac{1}{K_{eq}} = \frac{1}{2K} + \frac{1}{2K} \Rightarrow K_{eq} = K$

Branch ④ $\Rightarrow \frac{1}{K_{eq}} = \frac{1}{3K} + \frac{1}{3K} + \frac{1}{3K} \Rightarrow K_{eq} = K$

Branch ① and ② in parallel $\Rightarrow K_{eq} = 2K + K = 3K$ — ⑤

Branch ③ and ④ in parallel $\Rightarrow K_{eq} = K + K = 2K$ — ⑥

Branch ⑤ and ⑥ will be in parallel

$K_{eq_f} = 3K + 2K = 5K$

$T = 2\pi\sqrt{\frac{m}{K_{eq_f}}}$

$\boxed{T = 2\pi\sqrt{\frac{m}{5K}}}$

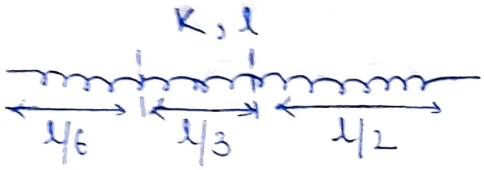
Note: If we have n-identical springs of $n.l = l$ and spring const $= k$ connected in series then $\boxed{K_{eq} = \frac{K}{n}}$

$$\frac{1}{K_{eq}} = \frac{1}{k} + \frac{1}{k} + \frac{1}{k} + \dots \text{ } n \text{ times} = \frac{n}{k}$$

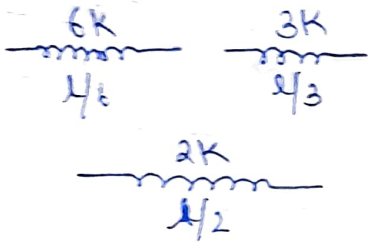
$$\boxed{K_{eq} = \frac{k}{n}}$$

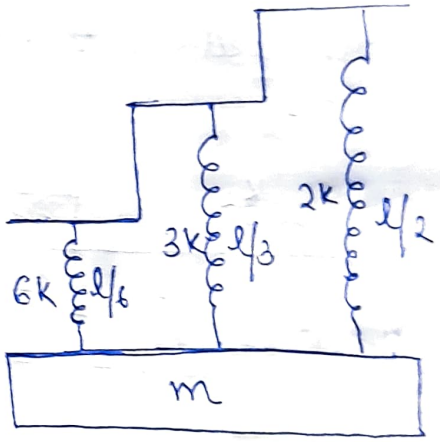
* Also we can say $\boxed{k \propto \frac{1}{\text{natural length of spring}}}$

Q



$K \propto \frac{1}{n.l}$



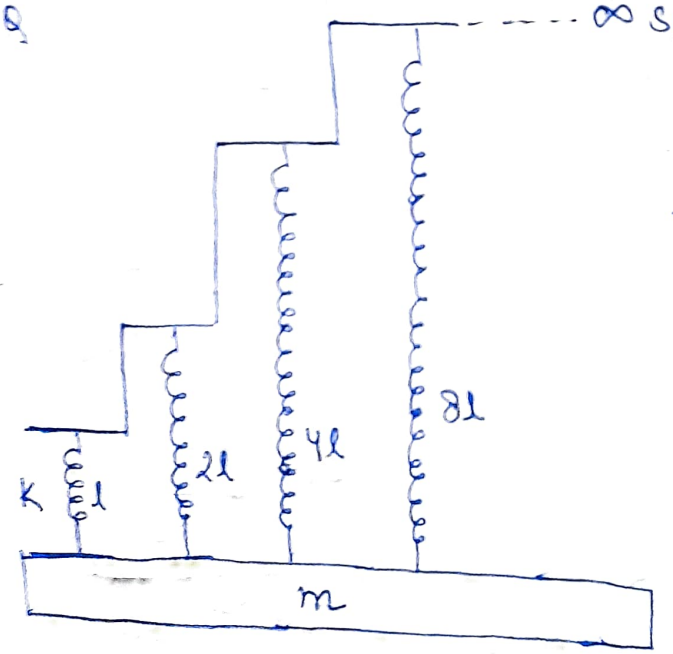


Spring is cut and arranged as shown Find time period of SHM of the block

$K_{eq} = 11K$

$$\boxed{T = 2\pi\sqrt{\frac{m}{11k}}}$$

Q



Find Time period for small oscillations of blocks

lengths → $l, 2l, 4l, 8l \dots$

⇒ Spring const → $K, \frac{K}{2}, \frac{K}{4}, \frac{K}{8} \dots$

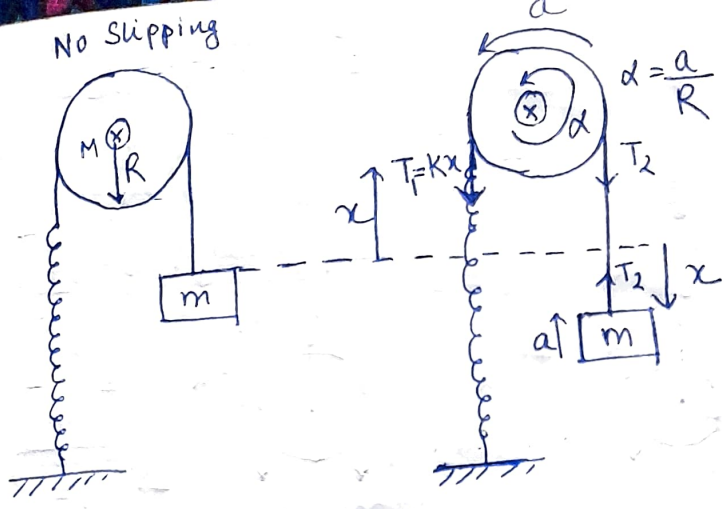
All are connected in parallel

$$K_{eq} = K\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \infty\right)$$

$$K_{eq} = K\left(\frac{1}{1 - \frac{1}{2}}\right) = 2K \quad \left\{S_\infty = \frac{a}{1-r}\right.$$

$$\boxed{T = 2\pi\sqrt{\frac{m}{2k}}}$$

Q No slipping



For small displacement in vertical direcⁿ prove that m is performing SHM. Also find time period of SHM.

$T_2 = ma$ — (1)

$+ \quad T_1R - T_2R = \frac{MR^2}{2} \cdot \frac{a}{R}$ — (2)

$T_1 = \left(m + \frac{M}{2}\right)a$

* In these type of problems we can neglect mg as it is a constant force and has no effect on T, ω, f

We know, $T_1 = kx$

$\Rightarrow \left(m + \frac{M}{2}\right)a = kx$

$$a = \left(\frac{2k}{2m+M}\right)x$$

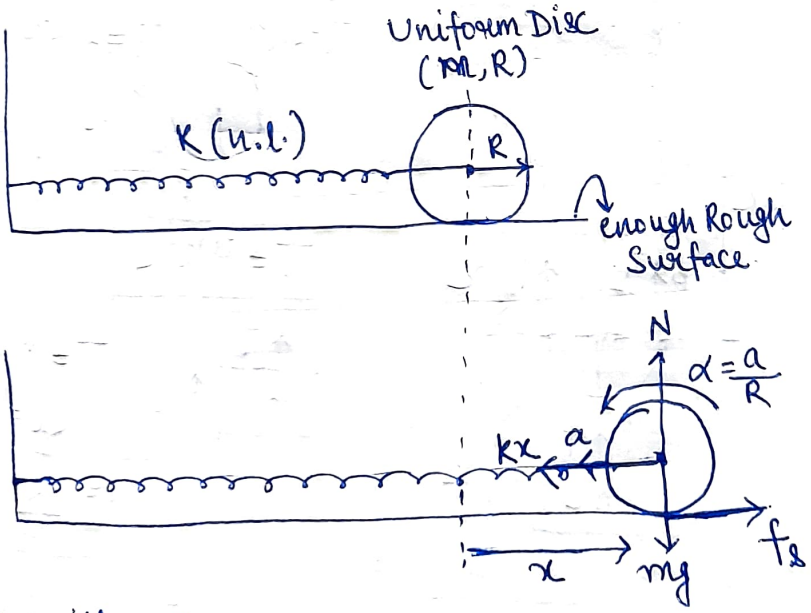
$$\omega = \sqrt{\frac{2k}{2m+M}}$$

$$T = 2\pi\sqrt{\frac{2m+M}{2k}}$$

As $a \propto x$ and F is opposite to x

$\Rightarrow$ SHM proved.

Q



For small displacement of disc in horizontal direⁿ. Prove that it will performs SHM. Also find T of SHM

$F_{ext} = m\vec{a}_{com}$

$kx - f_s = ma$ — (1)

$\tau_{com} = I_{com}\alpha$

$f_s R = \frac{mR^2}{2}\left(\frac{a}{R}\right)$

$f_s = \frac{ma}{2}$ — (2)

Put (2) in (1)

$kx = \frac{3}{2}ma$

$$a = \left(\frac{2k}{3m}\right)x$$

$$\omega = \sqrt{\frac{2k}{3m}}$$

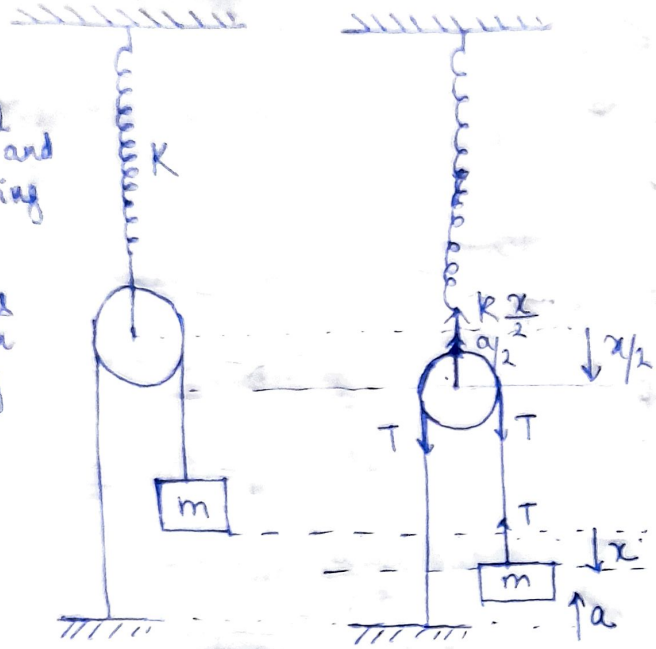
$$T = 2\pi\sqrt{\frac{3m}{2k}}$$

Disc will perform pure rolling due to torque of f_s and α will be developed in direcⁿ of a.

$\therefore$ As F opp to x and $a \propto x$

$\Rightarrow$ SHM proved.

Q

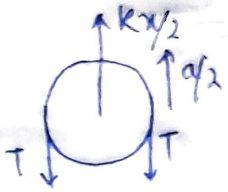


Find Time period for small oscillations of the block

Displacement of pulley: $x_p = \frac{0+x}{2}$

$x_p = x/2$

Acc^n of pulley: $a_p = \frac{0+a}{2} = \frac{a}{2}$

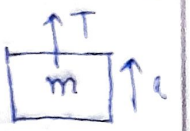


$m_p = 0$

$kx - 2T = 0$

$\frac{kx}{2} = 2T$

$\boxed{T = \frac{kx}{4}}$



$T = ma$

$\frac{kx}{4} = ma$

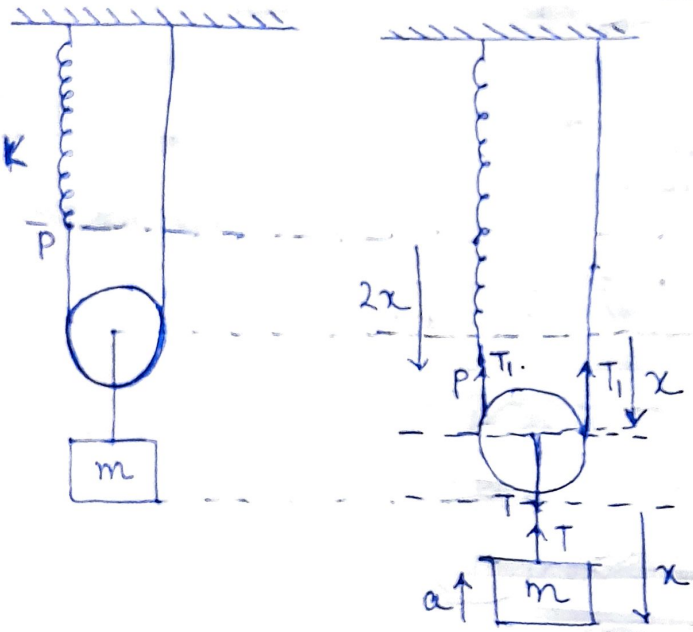
$\boxed{a = \left(\frac{k}{4m}\right)x}$

$\therefore$ SHM proved

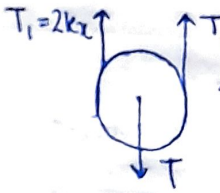
$\boxed{\omega = \sqrt{\frac{k}{4m}}}$

$\boxed{T = 2\pi\sqrt{\frac{4m}{k}}}$

Q

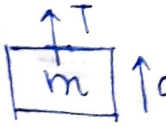


Find T for small oscillations of the block



$m_p = 0$

$\boxed{T = 4kx}$



$T = ma$

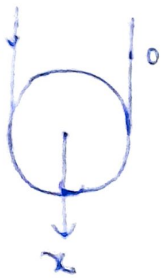
$4kx = ma$

$\boxed{a = \left(\frac{4k}{m}\right)x}$

$\boxed{\omega = \sqrt{\frac{4k}{m}}}$

$\boxed{T = 2\pi\sqrt{\frac{m}{4k}}}$

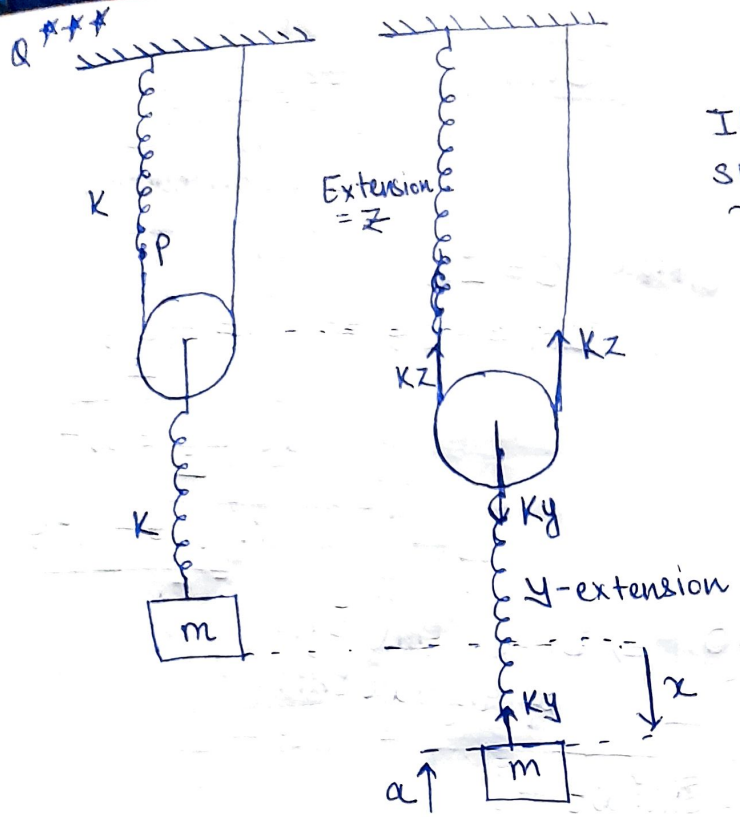
\#



Extension in spring (x_s):

$x = \frac{x_s + 0}{2} \Rightarrow \boxed{x_s = 2x}$

Q ***

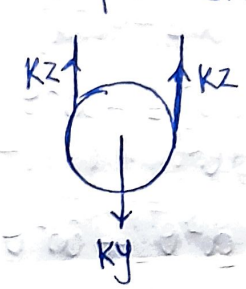


Find T for small oscillations of block.

If m comes down by x then spring will not have extension x as pulley will also come down.

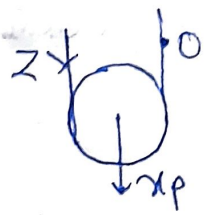
$F_{Restoring} = ky$

We need to find y in terms of x to prove SHM.



$2kz = ky$

$\boxed{y = 2z}$ — (1)



$x_p = \frac{z+0}{2}$

$x_p = \frac{z}{2}$

$\Rightarrow \boxed{y = x - \frac{z}{2}}$ — (2)

From (1) and (2)

$\boxed{y = \frac{y}{5}x}$

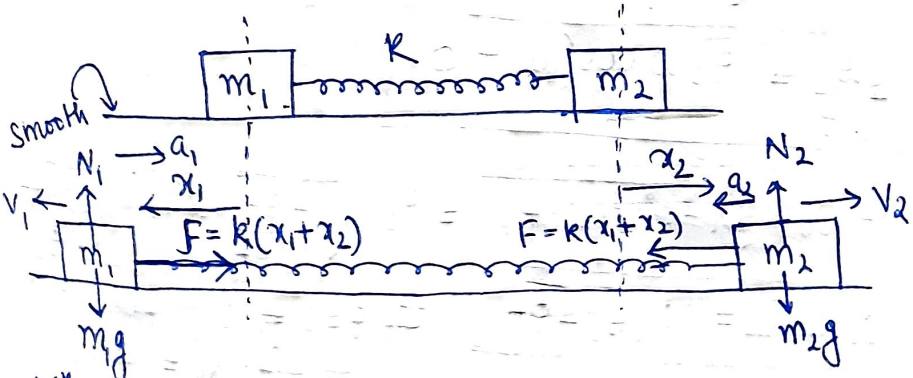
$F_{Restoring} = ky = \frac{4k}{5}x$

$ma = \frac{4k}{5}x \Rightarrow \boxed{a = \left(\frac{4k}{5m}\right)x}$

$\boxed{\omega = \sqrt{\frac{4k}{5m}}}$

$\boxed{T = 2\pi\sqrt{\frac{5m}{4k}}}$

Harmonic Oscillator:



We know,

$a_{com} = 0 \Rightarrow V_{com} = 0$

$\Rightarrow x_{com} = 0$

$\therefore \boxed{m_1x_1 = m_2x_2}$ — (1)

On block m_1

$F_{restoring} = k(x_1+x_2) = k\left(\frac{m_1x_1}{m_2} + x_1\right)$

$\Rightarrow \frac{k(m_1+m_2)x_1}{m_2}$

Now, $F = m_1a$

$\boxed{a = \left\{\frac{k(m_1+m_2)}{m_1m_2}\right\}x_1}$

$\boxed{\omega = \sqrt{\frac{k(m_1+m_2)}{m_1m_2}}}$

$\boxed{T = 2\pi\sqrt{\frac{m_1m_2}{k(m_1+m_2)}}}$

$\left(\mu = \frac{m_1m_2}{m_1+m_2} = \text{Reduced mass}\right)$

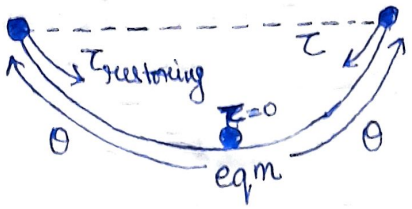
Angular SHM:

$\tau \propto \theta$

For angular SHM

$\vec{\tau} = -C\vec{\theta}$

C → SHM constant
-ve represents opp direction



Now, $\tau = I\alpha$

$\Rightarrow I\alpha = -C\theta \Rightarrow \boxed{\alpha = -\frac{C}{I}\theta}$

Let $\frac{C}{I} = \omega^2$

$\therefore \boxed{\alpha = -\omega^2\theta}$

Now, we know, $\alpha = -\omega^2\theta$

$\Rightarrow \frac{d^2\theta}{dt^2} = -\omega^2\theta$

$\Rightarrow \boxed{\frac{d^2\theta}{dt^2} + \omega^2\theta = 0}$ → Differential Eqn of Angular SHM.

Soln of this eqn $\Rightarrow \boxed{\theta = \theta_0 \sin(\omega t + \phi)}$

$\theta \rightarrow$ Angular displacement from eqm position
$\theta_0 \rightarrow$ Angular amplitude.
$\omega \rightarrow$ Angular Frequency
$\phi \rightarrow$ Phase constant / Initial phase
$\omega t + \phi \rightarrow$ Phase

Angular position of particle:

$$\boxed{\theta = \theta_0 \sin(\omega t + \phi)}$$

Angular Velocity of particle (Ω):

$$\boxed{\Omega = \frac{d\theta}{dt} = \theta_0 \omega \cos(\omega t + \phi)}$$

$$\Omega = \theta_0 \omega \cos(\omega t + \phi) = \theta_0 \omega \sqrt{1 - \sin^2(\omega t + \phi)} = \theta_0 \omega \sqrt{1 - \frac{\theta^2}{\theta_0^2}} = \theta_0 \omega \sqrt{\frac{\theta_0^2 - \theta^2}{\theta_0^2}}$$

$$\therefore \boxed{\Omega = \omega\sqrt{\theta_0^2 - \theta^2}}$$

If $\theta = 0 \Rightarrow$ eqm position
$\Rightarrow \Omega_{max} = \omega\theta_0$

$\boxed{\text{Maximum } \Omega = \theta_0\omega}$

Angular acceleration of particle:

$$\boxed{\alpha = \frac{d^2\theta}{dt^2} = -\theta_0\omega^2\sin(\omega t + \phi)}$$

$$\boxed{\alpha = -\omega^2\theta}$$

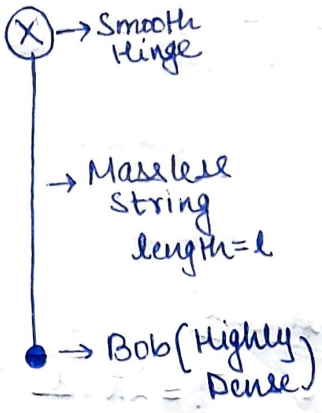
$$\boxed{\alpha_{max} = \omega^2\theta_0}$$

α_{max} will be at extreme position

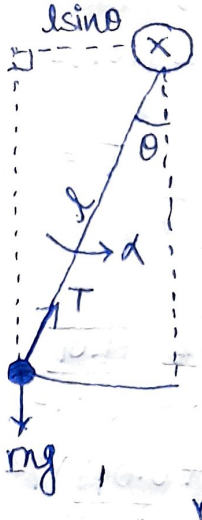
Simple Pendulum:

Note: θ must be small to perform SHM.

Vertical Plane



Bob should have more mass so that we can assume string to be massless in its comparison, but the volume should be least so air resistance can be neglected.



$\tau_{Res} = mgl\sin\theta$

If $\theta \to$ small, $\sin\theta \approx \theta$

$$\tau = mgl\theta$$

SHM proved.

Now, $\tau = I\alpha$

$mgl\theta = ml^2\alpha$

$$\alpha = \frac{g}{l}\theta$$

$$\omega = \sqrt{\frac{g}{l}}$$

$\therefore$ Time period of simple pendulum $= T = 2\pi\sqrt{\frac{l}{g}}$

Method 2: Since θ is small, we can approximate this motion as linear SHM to find T.

$F_{res} = mg\sin\theta$

Since $\theta \to$ small $\Rightarrow \sin\theta \approx \theta$

$F_{res} = mg\theta$

$ma = mg\theta = mg\left(\frac{x}{l}\right)$ {θ = Arc/Radius}

$$a = \frac{g}{l}(x) \Rightarrow \omega = \sqrt{\frac{g}{l}} \Rightarrow T = 2\pi\sqrt{\frac{l}{g}}$$

Q If l of simple pendulum increases by 21%. Find % change in T.

New $l = l' = l + \frac{21}{100}l = 1.21l$

Old $T = T = 2\pi\sqrt{\frac{l}{g}}$

New $T = T' = 2\pi\sqrt{\frac{1.21l}{g}}$

$= 1.1\left(2\pi\sqrt{\frac{l}{g}}\right) = 1.1(T)$

$\therefore T' = 1.1T$

% change $= \frac{T' - T}{T} \times 100$

$= \frac{1.1T - T}{T} \times 100$

% change in T = 10%

Q Find % change in T of simple pendulum, if l increases by 1% and g decreases by 2%.

$l' = l + \frac{l}{100} = 1.01\,l$ $\qquad$ $g' = g - \frac{2}{100}g = 0.98g$

$T = 2\pi\sqrt{\frac{l}{g}}$ $\qquad$ $T' = 2\pi\sqrt{\frac{1.01\,l}{0.98g}}$

Method1:

$$T' = 2\pi\sqrt{\frac{l}{g}}\sqrt{\frac{1.01}{0.98}}$$

$$T' = T\left(\frac{(1+0.01)}{(1-0.02)}\right)^{1/2}$$

$$T' = T(1+0.01)^{1/2}(1-0.02)^{-1/2}$$

we know $(1+x)^n = 1+nx$ if $x \ll 1$

$$T' = T\left(1+\frac{0.01}{2}\right)\left(1+\frac{0.02}{2}\right)$$

$$T' = T(1+0.005+0.01) = T(1.015)$$

$$\% \text{ change} = \frac{T'-T}{T}\times 100 = \frac{1.015T-T}{T}\times 100$$

% change in T = 1.5%

Method2:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T = 2\pi (l)^{1/2}(g)^{-1/2}$$

Take log and Differentiating both sides

$$\ln T = \ln 2\pi + \frac{1}{2}\ln l - \frac{1}{2}\ln g$$

$$\Rightarrow \frac{dT}{T} = \frac{1}{2}\left(\frac{dl}{l}\right) - \frac{1}{2}\left(\frac{dg}{g}\right)$$

As we have percentage.

$$\left(\frac{dT}{T}\times 100\right) = \frac{1}{2}\left(\frac{dl}{l}\times 100\right) - \frac{1}{2}\left(\frac{dg}{g}\times 100\right)$$

$\frac{dT}{T}\times 100 \Rightarrow$ % change in T

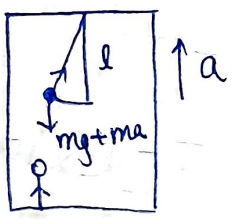
$\frac{dl}{l}\times 100 \Rightarrow$ % change in l

$\frac{dg}{g}\times 100 \Rightarrow$ % change in g

$$\Rightarrow \frac{dT}{T}\times 100 = \frac{1}{2}\times 1 - \frac{1}{2}(-2)$$

% change in T = 1.5%

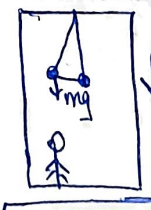
Ex



$$T = 2\pi\sqrt{\frac{l}{g_{eff}}} \qquad \therefore T = 2\pi\sqrt{\frac{l}{g+a}}$$

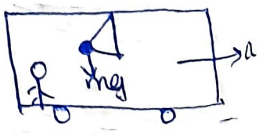
$$g_{eff} = |\vec{g} - \vec{a}|$$

Ex



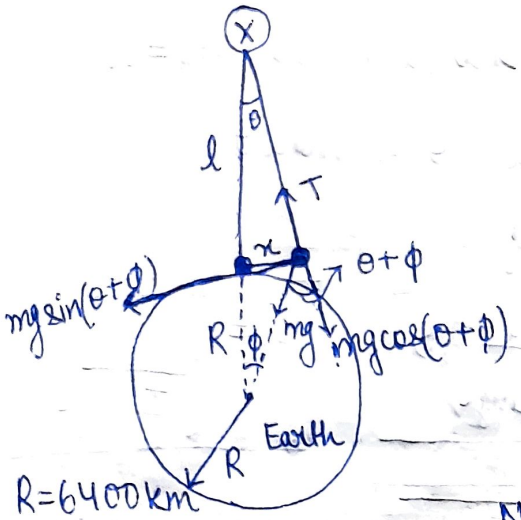
$$T = 2\pi\sqrt{\frac{l}{g-a}}$$

Ex $a \ll g$



$$T = 2\pi\sqrt{\frac{l}{\sqrt{g^2+a^2}}}$$

Long Pendulum:



$F_{res} = mg\sin(\theta+\phi)$

Since θ and ϕ are small

$\sin(\theta+\phi) \approx \theta+\phi$

$F_{res} = mg(\theta+\phi)$

$= mg\left(\frac{x}{l}+\frac{x}{R}\right)$

$$\boxed{F_{res} = mg\left(\frac{1}{l}+\frac{1}{R}\right)x} \Rightarrow \text{SHM Proved}$$

Now, $ma = mg\left(\frac{1}{l}+\frac{1}{R}\right)x$

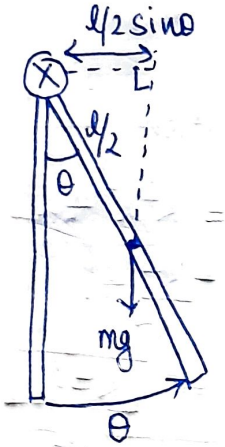
$$\boxed{a = g\left(\frac{1}{l}+\frac{1}{R}\right)x} \Rightarrow \boxed{\omega = \sqrt{g\left(\frac{1}{l}+\frac{1}{R}\right)}}$$

★ If we have infinitely long pendulum $\Rightarrow l \to \infty \Rightarrow \boxed{\omega = \sqrt{\frac{g}{R}}}$

$\therefore \boxed{T = 2\pi\sqrt{\frac{R}{g}}}$ $\boxed{T \approx 84.6 \text{ min}}$

Compound Pendulum / Physical Pendulum:

Ex



Given, Uniform Rod (m, l)

Prove that if rod is given small angular displacement, it will perform angular SHM.

Also find T of SHM.

$\tau = mg\frac{l}{2}\sin\theta$

Since $\theta \to$ small $\Rightarrow \sin\theta \approx \theta$

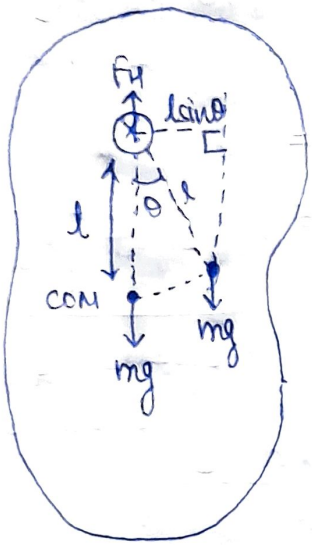
$$\boxed{\tau = \left(mg\frac{l}{2}\right)\theta} \Rightarrow \text{SHM proved}$$

Now, $\tau = I\alpha$

$mg\frac{l}{2}\theta = \frac{ml^2}{3}\alpha$

$$\boxed{\alpha = \left(\frac{3g}{2l}\right)\theta} \quad \boxed{\omega = \sqrt{\frac{3g}{2l}}} \quad \boxed{T = 2\pi\sqrt{\frac{2l}{3g}}}$$

★ If a body is hinged about any point except COM and made to oscillate, we call it a compound pendulum



★ Separation b/w Hinge and COM = l

$\tau_{res} = mgl\sin\theta \approx mgl\theta$

$I\alpha = mgl\theta$

$$\boxed{\alpha = \left(\frac{mgl}{I_{Hinge}}\right)\theta}$$

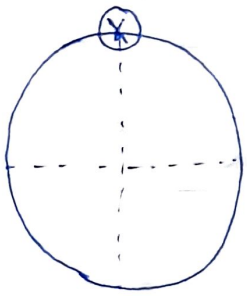
$$\boxed{\omega = \sqrt{\frac{mgl}{I_{Hinge}}}}$$

$$\boxed{T = 2\pi\sqrt{\frac{I_{Hinge}}{mgl}}}$$

where l is sep b/w Hinge and COM.

Q

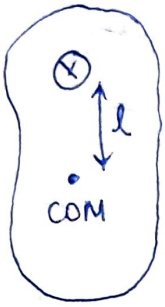
Uniform Disc (m, R)



Find Time period of oscillation.

$$T = 2\pi\sqrt{\frac{I_{Hinge}}{mgl}} = 2\pi\sqrt{\frac{\frac{3}{2}mR^2}{mgR}} = 2\pi\sqrt{\frac{3R}{2g}}$$

Note:



$$T = 2\pi\sqrt{\frac{I_{Hinge}}{mgl}}$$

$$= 2\pi\sqrt{\frac{I_{com} + ml^2}{mgl}}$$

$$= 2\pi\sqrt{\frac{mk^2 + ml^2}{mgl}}$$

$$\boxed{T = 2\pi\sqrt{\frac{\left(\frac{k^2}{l} + l\right)}{g}}}$$

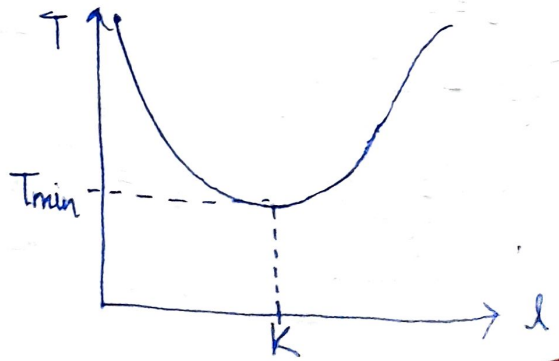
$k \rightarrow$ radius of gyration

Now, $f(l) = k^2 l^{-1} + l$

$f'(l) = -\frac{k^2}{l^2} + 1$

$\Rightarrow f'(l) = 0 \Rightarrow \boxed{k = l}$

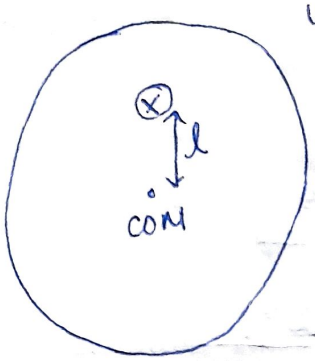
$$\boxed{\therefore T_{min} = 2\pi\sqrt{\frac{2k}{g}} \quad (\text{when } l = k)}$$



→ If we hinge near COM $\Rightarrow l \to 0, T \to \infty$

→ If we hinge far away from COM $\Rightarrow l \to \infty, T \to \infty$

Q



Uniform Disc (m, R)

From where this disc should be hinged so that T of its oscillation is least. Also find T_{min}.

$I_{com} = mK^2$

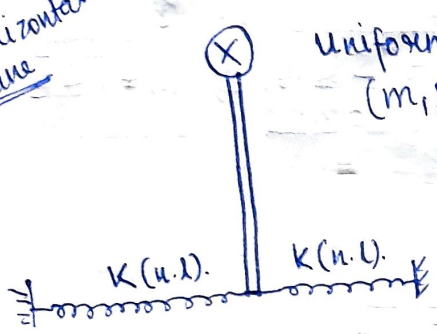
$\frac{mR^2}{2} = mK^2$

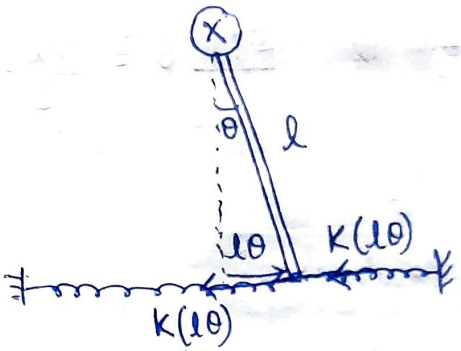
$\boxed{K = \frac{R}{\sqrt{2}}}$

$T_{min} = 2\pi\sqrt{\frac{2K}{g}}$

$\boxed{T_{min} = 2\pi\sqrt{\frac{\sqrt{2}R}{g}}}$

Q Horizontal Plane





For small angular displacements, prove that rod will perform angular SHM. Also find T of SHM.

$\boxed{\tau_{res} = (2kl\theta)l} \Rightarrow \tau \propto \theta$

$\Rightarrow$ SHM proved

Now $\tau = I\alpha$

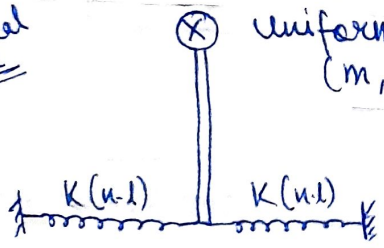
$2kl^2\theta = \frac{ml^2}{3}\alpha$

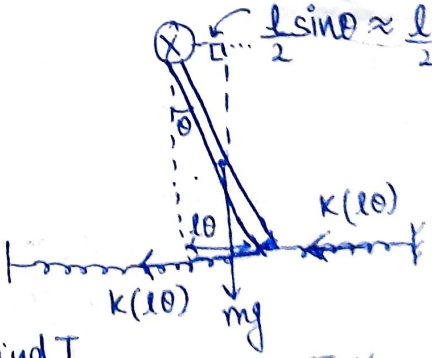
$\boxed{\alpha = \frac{6k}{m}\theta}$

$\boxed{\omega = \sqrt{\frac{6k}{m}}}$

$\boxed{T = 2\pi\sqrt{\frac{m}{6k}}}$

Q Vertical Plane





For small angular displacements prove that rod will perform angular SHM! Also find T.

$\boxed{\tau_{res} = (2kl\theta)l + mg\frac{l}{2}\theta} \rightarrow \tau \propto \theta$

SHM Proved

Now, $\tau = I\alpha$

$(2kl^2 + mg\frac{l}{2})\theta = \frac{ml^2}{3}\alpha$

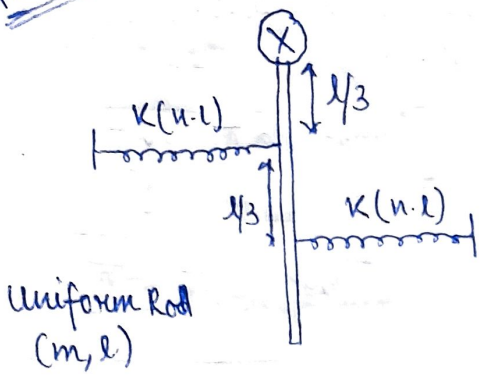
$\alpha = \frac{3(2kl^2 + mg\frac{l}{2})\theta}{ml^2}$

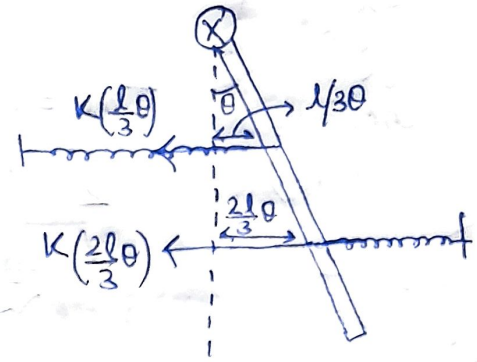
$\boxed{\alpha = \left(\frac{6K}{m} + \frac{3g}{2l}\right)\theta}$

$\boxed{\omega = \sqrt{\frac{6k}{m} + \frac{3g}{2l}}}$

$\boxed{T = \frac{2\pi}{\sqrt{\frac{6k}{m} + \frac{3g}{2l}}}}$

Q Smooth Horizontal Plane. For small angular displacements prove that rod performs angular SHM. Also find T for SHM.





$$\tau_{res} = K\frac{l}{3}\theta\left(\frac{l}{3}\right) + K\left(\frac{2l}{3}\right)\theta\left(\frac{2l}{3}\right) = \left(\frac{Kl^2}{9} + \frac{4Kl^2}{9}\right)\theta = \left(\frac{5kl^2}{9}\right)\theta$$

Now, $\tau = I\alpha \Rightarrow \frac{5kl^2}{9/3}\theta = \frac{ml^2}{3}\alpha \Rightarrow \boxed{\alpha = \left(\frac{5k}{3m}\right)\theta} \Rightarrow \boxed{\omega = \sqrt{\frac{5k}{3m}}}$

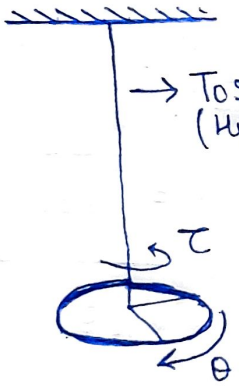
$$\boxed{T = 2\pi\sqrt{\frac{3m}{5k}}}$$

Torsional Pendulum: (Twist)

$$\boxed{\tau \propto \theta}$$

$$\boxed{\tau = C\theta}$$ {C = Torsional constant}

$\therefore$ SHM Proved



Now, $\tau = I\alpha$

$I\alpha = C\theta$

$$\boxed{\alpha = \frac{C}{I}\theta} \qquad \boxed{\omega = \sqrt{\frac{C}{I}}}$$

$$\boxed{T = 2\pi\sqrt{\frac{I}{C}}}$$

Superposition of SHM's:

We can superimpose two or more SHM's to find the resultant motion.

→ The resultant motion may or may not be a SHM.

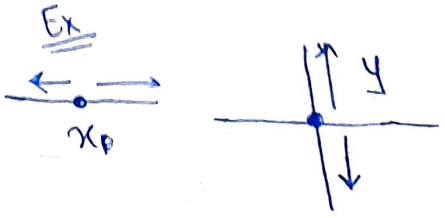
Ex

$x = x_1 + x_2$

↓ ↓ ↓

May or may not be SHM — SHM — SHM

Ex



$\vec{r} = x\hat{i} + y\hat{j}$

Ex Find Equation of resultant motion, if 2 SHM's are superimposed.

$x_1 = 4\sin\omega t \qquad x_2 = 3\cos\omega t$

$$x = x_1 + x_2 = 4\sin\omega t + 3\cos\omega t = 5\left(\frac{4}{5}\sin\omega t + \frac{3}{5}\cos\omega t\right)$$

$$= 5(\sin\omega t \cos 37^\circ + \cos\omega t \sin 37^\circ) = 5\sin(\omega t + 37^\circ)$$

$\therefore \boxed{x = 5\sin(\omega t + 37^\circ)}$ It is a SHM as it is of the form $x = A\sin(\omega t + \phi)$

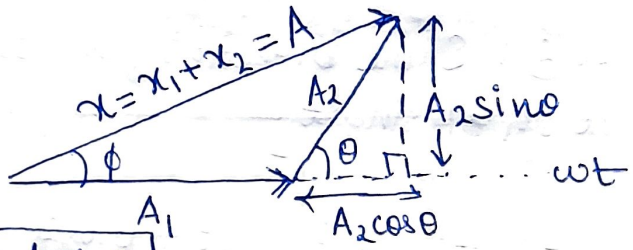
Case ①: Same Direction, same ω:-

Phasor: It is a vector representing sinosoidially varying eqn of SHM.

If we have $x = A\sin\omega t \Rightarrow$ Phasor → Magnitude = Amplitude (A)
→ Direction = Phase (ωt)

$\xrightarrow{A} \omega t$

Now, let $x_1 = A_1\sin\omega t$
$x_2 = A_2\sin(\omega t + \theta)$
} Superimpose and find resultant motion.



$x = x_1 + x_2$

$A = \sqrt{(A_1 + A_2\cos\theta)^2 + (A_2\sin\theta)^2}$

$\boxed{A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\theta}}$

$\boxed{\tan\phi = \frac{A_2\sin\theta}{A_1 + A_2\cos\theta}}$

$\therefore \boxed{x = A\sin(\omega t + \phi)}$

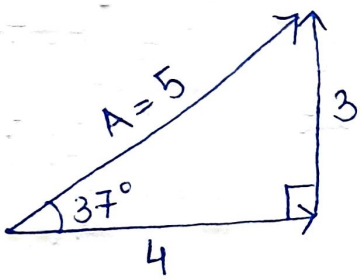
Q $x_1 = 4\sin\omega t$
$x_2 = 3\cos\omega t$
} If these two SHM's are superimposed, then find equation of resultant motion.

We have $x_1 = 4\sin\omega t \qquad x_2 = 3\cos\omega t = 3\sin(\omega t + \frac{\pi}{2})$

$x = x_1 + x_2$

$A_{res} = 5$

$\phi = 37^\circ$



$x_{res} = A\sin(\omega t + \phi)$

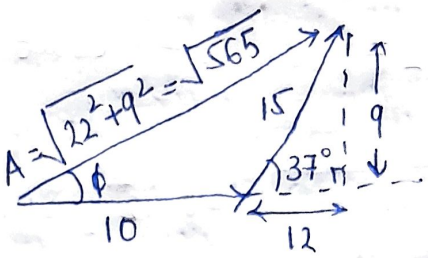
$\boxed{x = 5\sin(\omega t + 37^\circ)}$

Q $x_1 = 10\sin\omega t$
$x_2 = 15\sin(\omega t + 37°)$
} Find equation of resultant motion, if these 2 SHM's are superimposed.

$x = x_1 + x_2$

$\phi = \tan^{-1}\left(\frac{9}{22}\right)$

$$\boxed{x = \sqrt{565}\sin\left(\omega t + \tan^{-1}\left(\frac{9}{22}\right)\right)}$$



Q Find equation of resultant SHM if these 3 SHM's are superimposed

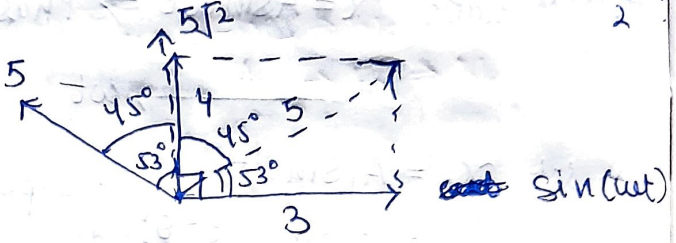
$x_1 = 3\sin\omega t$

$x_2 = 4\cos\omega t = 4\sin(\omega t + \pi/2)$

$x_3 = 5\cos(\omega t + 53°) = 5\sin(\omega t + \frac{\pi}{2} + 53°)$

$x = x_1 + x_2 + x_3$

$$\boxed{\begin{aligned} x &= 5\sqrt{2}\sin(\omega t + 98°) \\ &= 5\sqrt{2}\cos(\omega t + 8°) \end{aligned}}$$



Case (2): Same Direction, different ω :-

Let $x_1 = A\sin\omega t$

$x_2 = B\cos 2\omega t$

$x = x_1 + x_2 = A\sin\omega t + B\cos 2\omega t \neq$ SHM

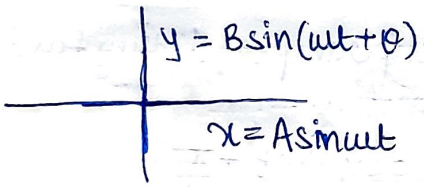
$x = A\sin\omega t + B\cos 2\omega t$

$\frac{d^2x}{dt^2} = -A\omega^2\sin\omega t - 4B\omega^2\cos 2\omega t$

$= -\omega^2\{A\sin\omega t + 4B\cos 2\omega t\}$

$\neq -\omega^2 x$ $\therefore$ It is not SHM.

Case (3): Mutually perpendicular direcn :-



$\vec{r} = x\hat{i} + y\hat{j}$

$\vec{r} = A\sin\omega t\,\hat{i} + B\sin(\omega t + \theta)\,\hat{j}$

(i) If $\theta = 0°$

$x = A\sin\omega t$

$y = B\sin\omega t$

$\frac{y}{x} = \frac{B}{A}$

$\Rightarrow \boxed{y = \frac{B}{A}x}$ → Straight line

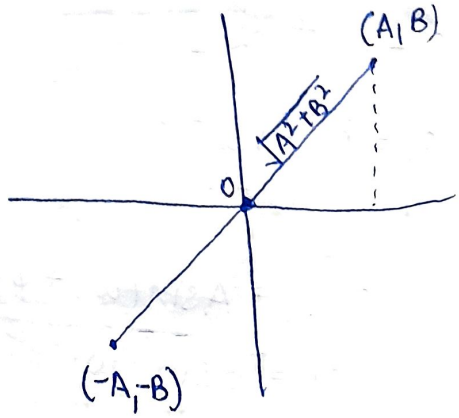
At $t = 0 \Rightarrow x = 0,\ y = 0$

At $t = \frac{\pi}{2\omega} \Rightarrow x = A,\ y = B$

At $t = \frac{\pi}{\omega} \Rightarrow x = 0,\ y = 0$

At $t = \frac{3\pi}{2\omega} \Rightarrow x = -A,\ y = -B$

At $t = \frac{2\pi}{\omega} \Rightarrow x = 0,\ y = 0$



(ii) If $\theta = \pi$

$x = A\sin\omega t$

$y = B\sin(\omega t + \pi) = -B\sin\omega t$

$\frac{y}{x} = \frac{-B}{A}$

$\Rightarrow \boxed{y = -\frac{B}{A}x}$

↳ Straight line

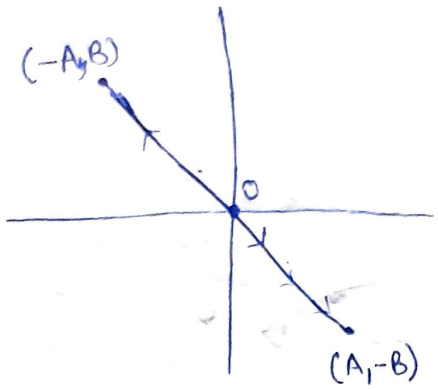
At $t=0 \Rightarrow x=0, y=0$

At $t=\frac{\pi}{2\omega} \Rightarrow x=A, y=-B$

At $t=\frac{\pi}{\omega} \Rightarrow x=0, y=0$

At $t=\frac{3\pi}{2\omega} \Rightarrow x=-A, y=B$

At $t=\frac{2\pi}{\omega} \Rightarrow x=0, y=0$



(iii) If $\theta = \frac{\pi}{2}$

$x = A\sin\omega t \Rightarrow \frac{x}{A} = \sin\omega t$ — (1)

$y = B\sin(\omega t + \frac{\pi}{2}) = B\cos\omega t$

$\Rightarrow \frac{y}{B} = \cos\omega t$ — (2)

(1)² + (2)²

$\boxed{\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1}$

↳ Ellipse

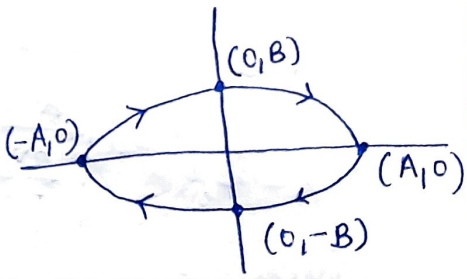
At $t=0 \Rightarrow x=0, y=B$

At $t=\frac{\pi}{2\omega} \Rightarrow x=A, y=0$

At $t=\frac{\pi}{\omega} \Rightarrow x=0, y=-B$

At $t=\frac{3\pi}{2\omega} \Rightarrow x=-A, y=0$

At $t=\frac{2\pi}{\omega} \Rightarrow x=0, y=B$



Q Find Equation of trajectory.

$x = 2\sin\omega t$

$y = 1\cos 2\omega t$

$y = 1\cos(2\omega t)$

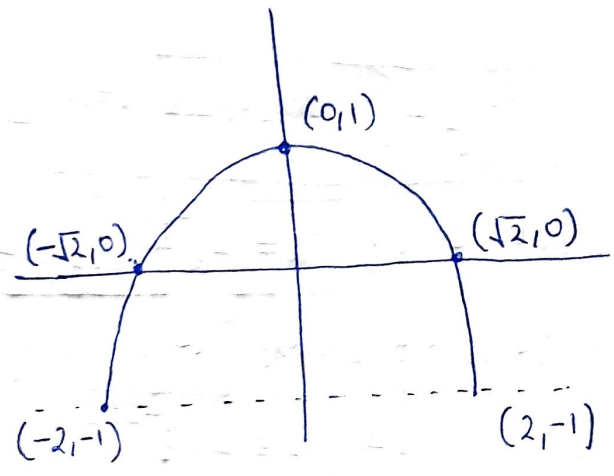
$= 1\{1 - 2\sin^2\omega t\}$

$= \{1 - 2\left(\frac{x^2}{4}\right)\}$

$\boxed{y = 1 - \frac{x^2}{2}} \Rightarrow$ Parabola

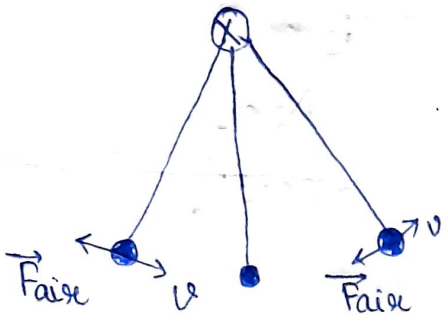
Also, $x = 2\sin\omega t \Rightarrow x \in [-2, 2]$

$y = \cos 2\omega t \Rightarrow y \in [-1, 1]$



Damped Oscillatory Motion:

→ In damped oscillatory motion, amplitude keeps decreasing.



Reasons of decreasing amplitude can be non-conservative forces.

$A\downarrow \Rightarrow$ Energy $\downarrow \Rightarrow$ Non-Conservative Force W.d $\Rightarrow$ -ve

Possible Non-conservative forces here,

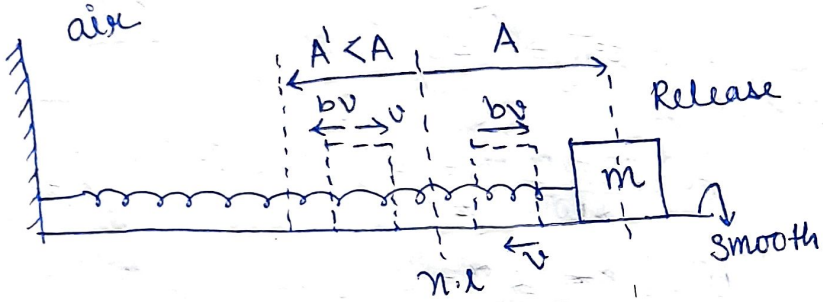
① Hinge can be rough

② $\vec{F}_{air}$ (air resistance) } Always acts opposite to v

★ $\boxed{\vec{F}_{air} \propto \vec{v}} \Rightarrow \boxed{\vec{F}_{air} = -b\vec{v}}$ (b = damping constant, Unit = kg/s)

Power of $\vec{F}_{air}$ = -ve $\Rightarrow$ W = -ve

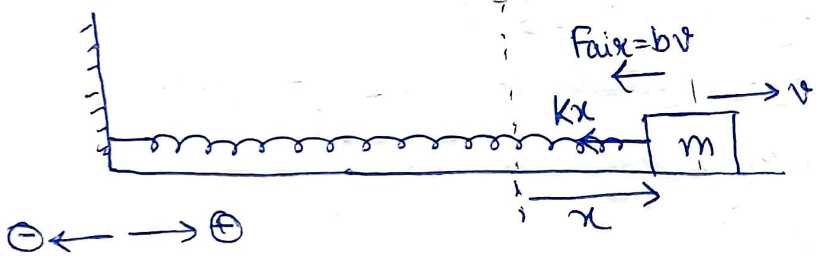
$\rightarrow$ E↓ $\Rightarrow$ A↓



$P = \vec{F} \cdot \vec{v}$

$P = -ve \Rightarrow W\,d = -ve$

$\Rightarrow E\downarrow \Rightarrow A\downarrow$



$-kx - bv = ma$

$\Rightarrow -kx - b\frac{dx}{dt} = m\frac{d^2x}{dt^2}$

$\Rightarrow m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$

$\therefore \boxed{\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = 0}$ ⇒ Differential Equation for damping.

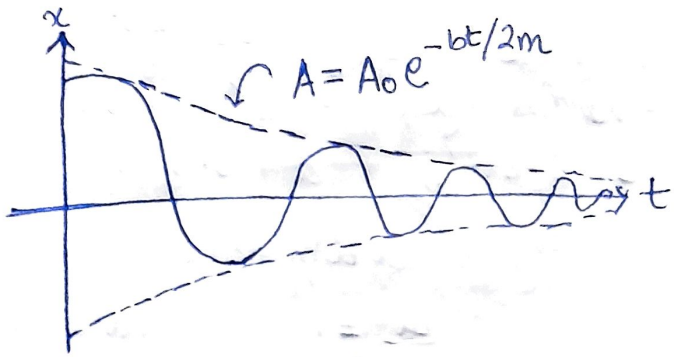
★ Soln of Eqn: $\boxed{x = Ae^{\frac{-bt}{2m}}\cos(\omega' t + \phi)}$

→ ω' is damped angular frequency $\boxed{\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}}$

→ Amplitude ⇒ $\boxed{A = A_0 e^{-bt/2m}}$ ∴ Amplitude is decreasing exponentially.

→ $\omega_0 = \sqrt{\frac{k}{m}}$ = Natural frequency. $\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$ = Damped Frequency

$\therefore \boxed{\omega' < \omega_0}$



Graph of x v/s t

$x = A_0 e^{\frac{-bt}{2m}} \cos(\omega' t + \phi)$

Energy of Oscillation: $E = \frac{1}{2} k A^2$

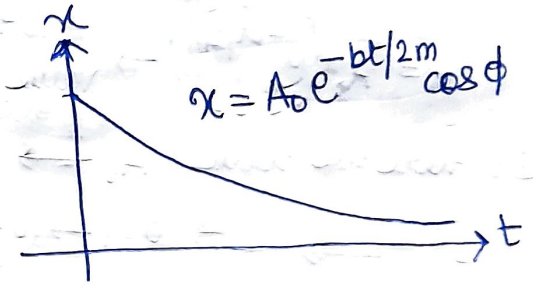
$= \frac{1}{2} k A_0^2 e^{-bt/m}$

$\therefore \boxed{E = E_0 e^{-bt/m}}$

Critical Damping: $\boxed{\omega' = 0}$

$\omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = 0 \Rightarrow \sqrt{\frac{4mk - b^2}{4m^2}} = 0 \Rightarrow \boxed{b^2 = 4mk}$

$\boxed{x = A_0 e^{-bt/2m} \cos\phi}$



Q Given $m = 200\,gm = 0.2\,kg$

$k = 90\,N/m$

$b = 40\,gm/s = 40 \times 10^{-3}\,kg/s$

Find ① Period of Oscillation

② Time taken for A to become half of initial

③ Time taken ~~for~~ Energy to drop to half of its initial value.

① $T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}} = \frac{2\pi}{\sqrt{\frac{90}{0.2} - \left(\frac{40\times10^{-3}}{2\times200\times10^{-3}}\right)^2}}$

$= \frac{2\pi}{\sqrt{450 - \left(\frac{1}{10}\right)^2}} = \frac{2\pi}{\sqrt{450 - 0.01}} = \frac{2\pi}{\sqrt{449.99}}\,s$

② $A = A_0 e^{-bt/2m}$

$\frac{A_0}{2} = A_0 e^{-bt/2m}$

$\frac{1}{2} = e^{-bt/2m}$

Taking ln both sides

$\ln\frac{1}{2} = -\frac{bt}{2m}$

$\Rightarrow \ln 2 = \frac{bt}{2m}$

$t = \frac{2m}{b} \ln 2$

$t = \frac{2\times0.2}{40\times10^{-3}} \times 0.693$

$\boxed{t = 6.93\,s}$

③ $E = E_0 e^{-bt/m}$

$\frac{E_0}{2} = E_0 e^{-bt/m}$

$\frac{1}{2} = e^{-bt/m}$

Taking ln both sides

$\ln 2 = \frac{+bt}{m}$

$t = \frac{m}{b} \ln 2$

$\boxed{t = \frac{6.93}{2}\,s}$

Given Simple pendulum of mass m. Also $F_{air} = -bv$. to Average life time is given as the time taken by A to reduce 1/e times its original value. Find average life time of pendulum

a) 1/b b) 2/b c) 0.693/b d) 0.633/b

$A = A_0 e^{-bt/2m}$

$\frac{A_0}{e} = A_0 e^{-bt/2m}$

$e^{-1} = e^{-bt/2m}$

Taking ln both sides

$\frac{bt}{2m} = 1$

$\boxed{t = \frac{2m}{b}}$

→ Now there is no option as 2m/b but sometimes Fair is taken as $-\frac{b}{m}v$ due to which a factor of m gets missing. So in exam if answer and options just misses by a 'm', we can tick by neglecting m.

Q It takes 5s for A_0 to become $0.9A_0$. How much time will it take for A_0 to become $0.729A_0$.

$A = A_0 (e^{-bt/2m})$ → A noteable pattern for $e^{-bt/2m}$ is that its value changes as $1 \to x \to x^2 \to x^3 \to x^4 \to \dots$ in equal time intervals.

So here we have $A_0 \longrightarrow 0.9A_0$

$\Rightarrow e^{-bt/2m} \Longrightarrow 1 \to 0.9 \to (0.9)^2 \to (0.9)^3$

$\Downarrow \quad \Downarrow \quad \Downarrow$

$0.81 \quad \& 0.729$

∴ Time taken $\Rightarrow 5+5+5 = 15s$.

∴ It takes 15s for A_0 to become $0.729A_0$

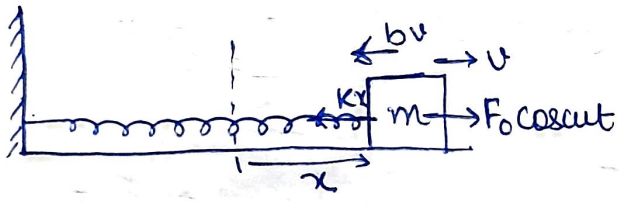
Forced Oscillations: occurs when system is driven by period force external to it.

For forced oscillations, we have to do some work (W_{ext})

If $W_{ext} > |W_{Air}| \Rightarrow E\uparrow, A\uparrow$ {W_{ext} = +ve}

If $W_{ext} = |W_{Air}| \Rightarrow \Delta E = 0$

E = const, A = const



$F_0 \cos\omega t - kx - bv = ma$

$\boxed{\frac{d^2x}{dt^2} + \frac{b}{m}\frac{dx}{dt} + \frac{k}{m}x = F_0 \cos\omega t}$

System's natural frequency = ω_0

External Agents frequency = ω

For resonance, $\boxed{\omega = \omega_0}$

→ A is max^m ~~if force is small~~

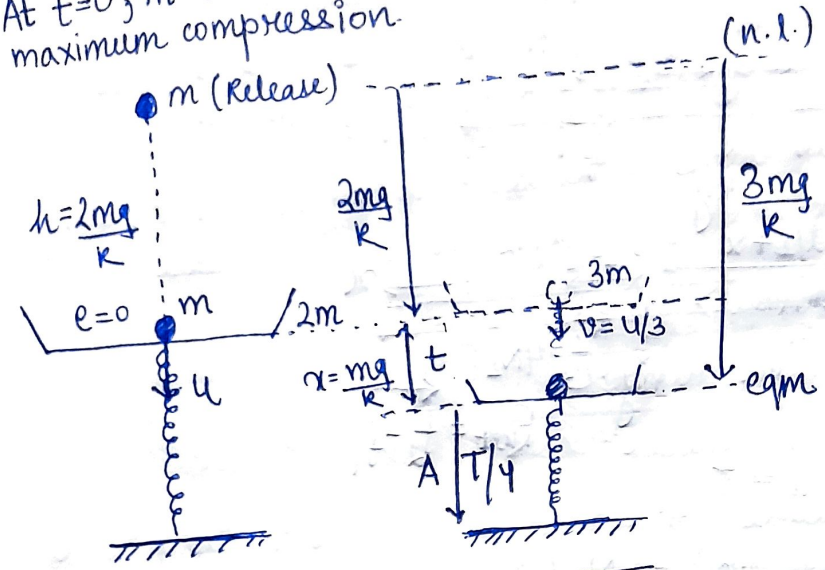
$$\boxed{A = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + \left(\frac{b\omega}{m}\right)^2}}}$$

Q For small damping $A \propto$?

For small damping $b \to 0$

$$A = \frac{F_0/m}{\sqrt{(\omega^2-\omega_0^2)^2}} = \frac{F_0/m}{\omega^2-\omega_0^2} \quad \therefore A \propto \frac{1}{\omega^2-\omega_0^2}$$

Q At $t=0$, m sticks to $2m$, then find time at which spring will have maximum compression.



$$u = \sqrt{2gh} = \sqrt{2g\left(\frac{2mg}{k}\right)}$$

$$\omega = \sqrt{\frac{k}{3m}}$$

M/c before & after collision:

$$m\sqrt{2g\left(\frac{2mg}{k}\right)} + 0 = 3mv$$

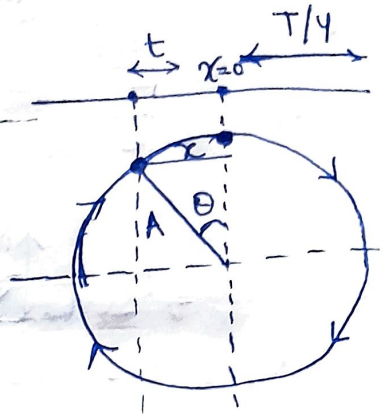
$$\boxed{v = \frac{u}{3}}$$

$$T = 2\pi\sqrt{\frac{3m}{k}}$$

Now, $v = \omega\sqrt{A^2-x^2}$

$$\Rightarrow \frac{2g\sqrt{\frac{m}{k}}}{3} = \sqrt{\frac{k}{3m}}\sqrt{A^2 - \frac{m^2g^2}{k^2}}$$

$$\Rightarrow \boxed{A = \sqrt{\frac{7}{3}}\frac{mg}{k}}$$



$$\sin\theta = \frac{x}{A}$$

$$\theta = \sin^{-1}\left(\frac{x}{A}\right) = \omega t$$

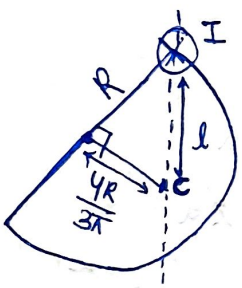
$$= \sin^{-1}\left(\frac{mgk}{k\sqrt{\frac{7}{3}}mg}\right) = \sqrt{\frac{k}{3m}}\,t$$

$$\boxed{t = \sqrt{\frac{3m}{k}}\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)}$$

Time for max^m comp. $= t + \frac{T}{4}$

$$\Rightarrow \boxed{\sqrt{\frac{3m}{k}}\sin^{-1}\left(\sqrt{\frac{3}{7}}\right) + \frac{\pi}{2}\sqrt{\frac{3m}{k}}}$$

Q

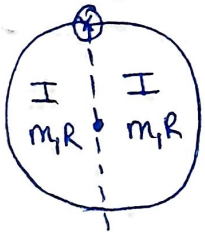


Uniform Half Disc (m, R)

Find Time period of small angular oscillations of this half disc.

$$l = \sqrt{R^2 + \frac{16R^2}{9\pi^2}}$$

$$\boxed{l = \frac{R}{3\pi}\sqrt{16+9\pi^2}}$$



MOI about given axis:

$$2I = \frac{3}{2}(2m)R^2$$

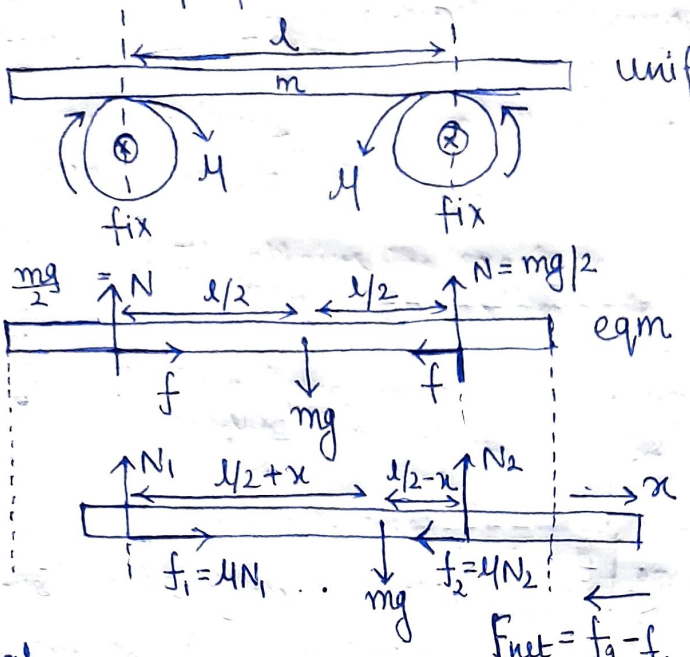
$$\boxed{I = \frac{3}{2}mR^2}$$

$$T = 2\pi\sqrt{\frac{I_H}{mgl}}$$

$$T = 2\pi\sqrt{\frac{3/2\ mR^2}{mg\frac{R}{3\pi}\sqrt{16+9\pi^2}}}$$

$$\boxed{T = 2\pi\sqrt{\frac{9\pi R}{2g\sqrt{16+9\pi^2}}}}$$

Q Prove that if planck is displaced slightly along horizontal direction, it will perform SHM. Also find T.



Rotational Forc equilibrium, $\tau_{N_1} = \tau_{N_2}$

$$N_1\left(\frac{l}{2}+x\right) = N_2\left(\frac{l}{2}-x\right) \quad \text{—①}$$

Also $N_1 + N_2 = mg$ —②

$$\Rightarrow N_1 + N_1\frac{(l+2x)}{(l-2x)} = mg \Rightarrow N_1\left\{\frac{2l}{l-2x}\right\} = mg$$

$$\boxed{N_1 = \frac{mg}{2l}(l-2x)} \qquad \boxed{N_2 = \frac{mg}{2l}(l+2x)}$$

$$F_{restoring} = f_2 - f_1 = \mu N_2 - \mu N_1 = \mu(N_2 - N_1)$$

$$= \mu\left\{\frac{mg}{2l}4x\right\}$$

$$\boxed{F_{res} = \frac{2\mu mg x}{l}} \quad F \propto x \Rightarrow \text{SHM proved}$$

Now, $F = ma$

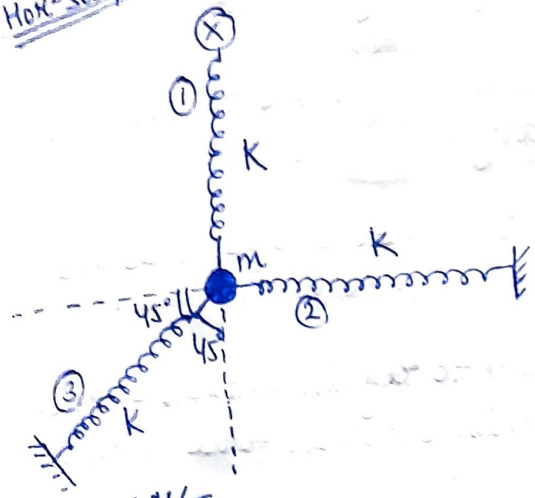
$$\frac{2\mu mg x}{l} = m a$$

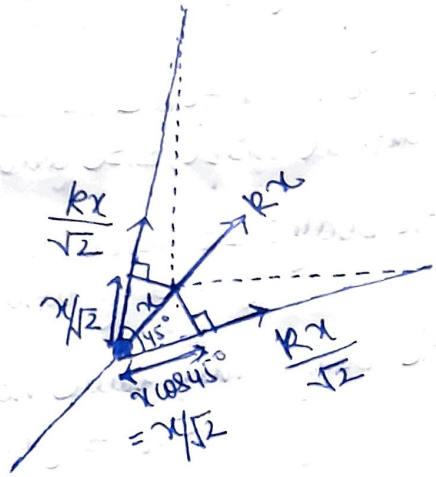
$$\boxed{a = \frac{2\mu g}{l}x} \qquad \boxed{\omega = \sqrt{\frac{2\mu g}{l}}}$$

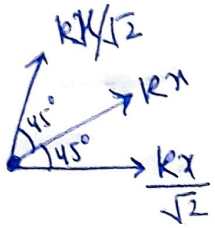
$$\boxed{T = 2\pi\sqrt{\frac{l}{2\mu g}}}$$

Q Mass m is displaced slightly to compress spring ③. Find T.

Smooth Horizontal Surface





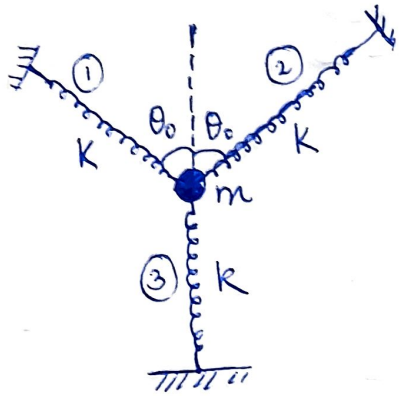


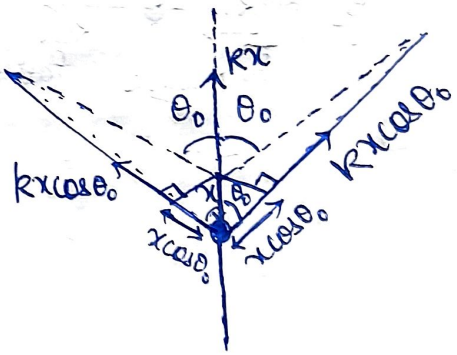
$$F_{res} = kx + \frac{kx}{\sqrt{2}}\cos 45° + \frac{kx}{\sqrt{2}}\cos 45° = 2kx$$

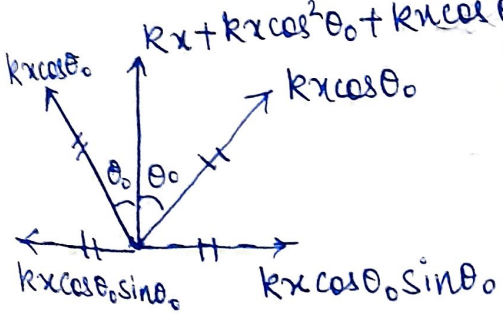
Now, $2kx = ma \Rightarrow \boxed{a = \left(\frac{2k}{m}\right)x}$

$$\boxed{T = 2\pi\sqrt{\frac{m}{2k}}}$$

Q Mass m is displaced slightly to compress spring ③. Find T







$$\boxed{F_{res} = (k + 2k\cos^2\theta_0)x}$$

$$(k + 2k\cos^2\theta_0)x = ma$$

$$\boxed{a = \frac{k(1+2\cos^2\theta_0)x}{m}}$$

$$\boxed{\omega = \sqrt{\frac{k(1+2\cos^2\theta_0)}{m}}}$$

$$\boxed{T = 2\pi\sqrt{\frac{m}{k(1+2\cos^2\theta_0)}}}$$